UNISON: Unified Hand Interaction State Generation

Appendix

This appendix provides additional details on dataset construction, model architecture, training and evaluation protocols, and further experimental analyses to complement the main paper. Additional visualizations and resources are available on our project website at https://unison-page.github.io.

A. Extended Related Work

This appendix presents an extended discussion of related work. It provides a more detailed review of 3D hand pose estimation, hand-object contact modeling, vision-based hand pressure estimation, and generative modeling of dexterous hand states.

A.1. 3D Hand Pose Estimation

Recovering 3D hand geometry from monocular RGB observations is a fundamental step toward understanding hand-object interaction. Existing methods have progressed from direct regression with parametric hand models to transformer-based reconstruction and temporally coherent estimation.

Regression and Transformer-Based Estimation. Early methods primarily rely on regression and parametric model integration. FrankMocap [34], for example, estimates articulated 3D hand motion from in-the-wild RGB images within a unified body reconstruction pipeline. Subsequent methods introduce transformer architectures to model dependencies across hand joints and mesh vertices. METRO [21] directly reconstructs 3D joints and mesh vertices with transformer encoders, while Mesh Graphormer [22] integrates graph convolution into the transformer architecture to preserve local mesh topology and capture global geometric dependencies. More recent methods further improve reconstruction robustness through large-scale training and stronger image-level priors. HaMeR [28] achieves reliable hand mesh recovery across diverse in-the-wild images, and WiLoR [30] strengthens monocular reconstruction under challenging viewpoints, scales, and visual conditions.

Occlusion-Aware and Temporal Estimation. Hand-object interaction creates additional difficulties because the manipulated object frequently occludes fingers, palm regions, and other geometric cues. HandOccNet [27] improves occlusion-robust hand mesh estimation by propagating visible hand evidence to occluded regions through transformer-based feature processing. UniHOPE [38] further models the relationship between hand geometry and object interaction. It combines grasp-aware feature fusion with occlusion-aware training to support both isolated-hand and hand-object pose estimation within a unified framework. Beyond single-image reconstruction, HaPTIC [45] estimates 4D hand trajectories from monocular videos and exploits long-range temporal information to maintain coherent hand motion during interaction. These developments extend hand pose estimation toward occlusion-aware and temporally consistent hand-object understanding.

However, hand pose specifies only the geometric configuration of an interaction. Similar hand configurations may produce different contact regions and pressure distributions depending on object properties, surface correspondence, and interaction stage. UNISON therefore treats pose as one component of a unified hand interaction state and models it jointly with hand-surface contact and pressure over time.

A.2. Hand-Object Contact Modeling

Hand-object contact identifies where physical engagement occurs and provides geometric constraints for plausible hand-object configurations. Existing studies model contact either as an optimization cue for pose refinement or as a prediction target inferred from visual, geometric, and semantic inputs.

Geometry-Based Contact Modeling. ContactOpt [8] predicts target surface contacts from initialized hand and object meshes and refines the hand pose through differentiable contact objectives. CPF [44] represents contact with a potential field and combines learned predictions with model fitting to reduce separation and interpenetration. These methods establish contact as a structured geometric variable that complements visual pose estimation.

Contact Prediction and Generation. Subsequent studies extend contact modeling toward direct estimation and controllable generation. S^2Contact [37] learns 3D contact from monocular images through a semi-supervised graph-based framework that exploits visual and geometric consistency. NL2Contact [51] introduces language-guided contact generation and uses staged diffusion models to produce controllable 3D contact patterns. More recently, HACO [14] addresses class and spatial imbalance when learning dense hand-surface contact across diverse interaction scenarios. ContactPrompt [15] further uses multimodal large language models and structured hand-geometry prompts for training-free dense contact estimation.

Existing methods primarily use contact to refine hand-object geometry or predict contact distributions from visual and geometric observations. These studies establish contact as an important representation for interaction understanding, while its temporal evolution and relationship with tac-

tile load remain less explored. UNISON extends this direction by representing contact on a fixed hand-surface taxel topology and generating it jointly with pose and pressure as a temporally coherent interaction-state sequence.

A.3. Vision-Based Hand Pressure Estimation

Pressure describes how tactile load is distributed over the hand surface and provides physical information beyond binary contact. Vision-based pressure estimation seeks to recover this hidden signal from visual appearance, hand motion, and interaction context.

Image-Based Pressure Estimation. PressureVision [9] estimates hand pressure from a single RGB image using visual cues such as soft-tissue deformation, hand configuration, skin color variation, and local shadows around contact regions. PressureVision++ [10] extends this formulation to more diverse fingertip-pressure scenarios and uses weak prompting signals to scale pressure supervision. EgoPressure [54] further studies pressure estimation from egocentric RGB images with optional 3D hand-pose guidance. It represents pressure in the MANO [33] UV domain and maps the prediction onto the 3D hand surface.

Video-Based Pressure Estimation. Recent methods extend vision-based pressure estimation to videos and more natural manipulation scenarios. TouchAnything [56] introduces a multi-view vision-to-touch framework that predicts dense bimanual tactile fields from egocentric and wrist-mounted videos. EgoTactile [48] focuses on everyday grasping and estimates full-hand pressure from egocentric videos using pressure-glove supervision. These methods exploit temporal observations to infer dense tactile states during complex manipulation.

Existing methods demonstrate that hand pressure can be inferred from visual observations and progressively extend pressure estimation from single images to temporally informed video settings. However, pressure is commonly modeled as an independent prediction target, with limited consideration of its dependency on hand geometry and contact formation. UNISON extends this direction by representing pressure and contact on a shared hand-surface taxel topology and jointly generating pose, contact, and pressure as a temporally coherent interaction-state sequence.

A.4. Generative Modeling of Dexterous Hand States

Generative models provide a flexible approach to representing dexterous hand states because the same object and task conditions may admit multiple feasible hand configurations. Existing studies mainly use diffusion or autoregressive models to synthesize robotic grasps and contact patterns from object geometry, affordances, and task conditions.

Contact-Aware Grasp Synthesis. GenDexGrasp [20] uses a hand-agnostic contact representation to synthesize diverse grasps across different multi-finger robotic hands. GrainGrasp [53] predicts fingertip-specific contact maps that provide fine-grained guidance for grasp optimization. Diffusion-based methods further improve the diversity and flexibility of grasp generation. UGG [24] jointly models object point clouds, dexterous hand parameters, and contact points within a unified diffusion architecture. DexDiffuser [39] generates grasps from partial object point clouds and combines diffusion sampling with grasp evaluation and refinement.

Controllable and Cross-Embodiment Grasping. Recent studies place greater emphasis on functional control, physical feasibility, and generalization across objects and hand embodiments. DexGrasp-Diffusion [52] learns a unified diffusion model for multiple dexterous hand embodiments and generates functional grasps conditioned on object affordances. DexGrasp Anything [55] incorporates physical constraints into diffusion training and sampling to improve grasp feasibility over diverse objects. G-DexGrasp [13] uses retrieved contact and affordance priors to guide grasp generation for unseen object categories and language-based task instructions. UniMorphGrasp [40] conditions diffusion on structured hand morphology and object geometry to support cross-embodiment generation, including transfer to unseen hand structures. DextER [18] autoregressively generates embodiment-aware contact tokens followed by dexterous hand configurations, enabling language-guided and controllable grasp synthesis.

These methods demonstrate the effectiveness of generative modeling for diverse and controllable dexterous grasp synthesis. Their outputs mainly describe grasp configurations or geometric contact relationships at a selected interaction stage. UNISON parameterizes a video-conditioned flow over complete interaction-state trajectories and jointly generates articulated pose, taxel-level contact, and pressure throughout the interaction.

B. Additional Details on UNISON-Sim

B.1. Simulation Protocol

Episode Construction. UNISON-Sim starts from BODex [5] tabletop grasp records for a UR10e arm equipped with a Shadow Hand. Each record provides the object asset, object scale, object pose, and staged hand-arm configurations. We replay these records in MuJoCo and extend the original grasp-and-lift motion with a hold phase and a return motion. The exported phases are approach, pregrasp, grasp, squeeze, lift, hold-lift, return-squeeze, return-grasp, return-pregrasp, and reverse-approach. An episode therefore covers contact formation, load bearing, release, and final clearance. To make data construction explicit, an episode denotes one complete simulated rollout and a training sample denotes a single-view temporal clip

extracted from a retained episode. Candidate episodes are retained only when the replay satisfies four success checks. The object must be lifted, the hand must form valid contact with the object, the return motion must place the object back within the accepted height range, and the hand must clear the object after release. Episodes with missing metadata or insufficient frames are rejected before training manifest construction.

Physical Parameter Assignment. Each episode is instantiated with object-specific physical properties. Object mass is taken from the scaled BODex metadata when the recorded value is physically plausible. Otherwise, it is estimated from the object dimensions, category, material density, and an approximate fill factor. Material and contact properties, including friction and restitution, are assigned according to the object category and material. The resolved physical attributes, together with their provenance, are stored in the episode metadata and used consistently during simulation and context construction.

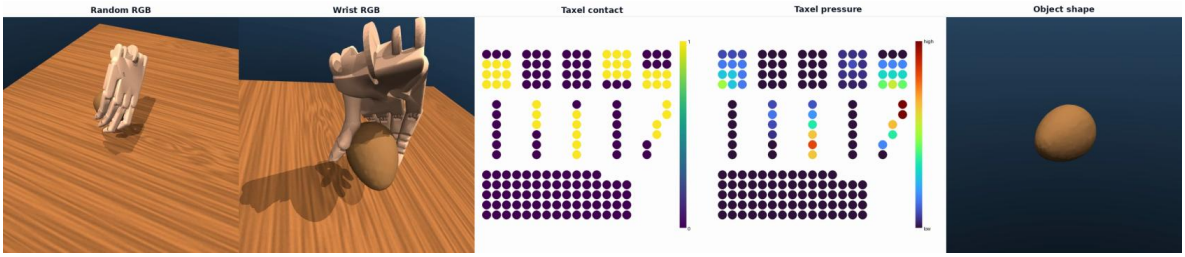
Tactile Assignment. Contact and pressure are exported on the same 162-taxel topology used in the unified state representation. The layout is a fixed 17×19 hand-surface grid with inactive cells removed. Taxels are grouped into fingertips, middle phalanges, proximal phalanges, and the palm. The five fingertips contain 12 taxels each, each middle and proximal phalanx contains 3 taxels, and the palm contains 72 taxels. The topology graph uses grid neighbors and finger-chain edges, yielding 431 non-self edges plus one self-loop per taxel in the model adjacency. At each frame, MuJoCo hand-object contacts are matched to the corresponding hand region and assigned to valid taxel anchors in that region. For taxel i at frame t, let $f_{t,i}$ denote the accumulated non-negative normal contact force assigned to that taxel, measured in N. The binary contact label is $c_{t,i} = 1$ when $f_{t,i} > 0$, and the pressure proxy used as the default training target is $r_{t,i} = \log(1 + f_{t,i})$ before dataset-level scaling. At evaluation and export time, the denormalized model prediction $\hat{r}_{t,i}$ is mapped back to the force domain as $\hat{f}_{t,i} = \exp(\hat{r}_{t,i}) - 1$, where $\hat{r}_{t,i}$ denotes the predicted log-pressure proxy and $\hat{f}_{t,i}$ denotes the reported predicted normal force. The same taxel index i is used for contact and pressure, so the two tactile fields remain pointwise aligned. The logarithm is applied to the numerical force value expressed in N. Logarithmic transformation and dataset scaling define the training representation, while inverse-transformed forces are expressed in N.

RGB Rendering. The current release exports 480×480 RGB videos at 15 fps with 143 frames per episode. By default, two views are retained. The random oblique view is a third-person camera sampled around the interaction with distance in $[0.38, 0.68]$ m, azimuth in $[0°, 360°)$, height offset in $[0.22, 0.40]$ m, and small target jitter. The wrist view is a forearm-mounted camera with a $95°$ field of view. The benchmark uses the random view stream as the visual input, while the wrist stream is kept for future embodied-view experiments. Rendering randomization changes appearance while keeping the simulator state that produces labels fixed. The tabletop style is sampled from texture and solid-color presets, and scene lighting is sampled from soft key, top, and fill lights with position, direction, diffuse, and ambient perturbations. During rendering, robot geometries outside the hand-object interaction are made transparent, while the hand, object, and tabletop remain visible. This keeps the visual input centered on the hand-object interaction. Figure A1 visualizes the representative samples of UNISON-Sim.

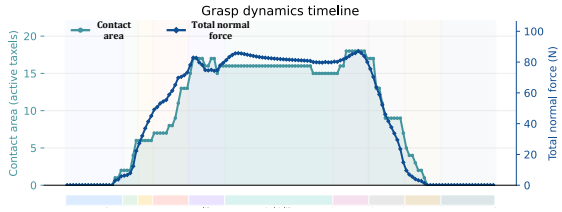
Object Context. Each retained episode provides object context through three complementary representations. The text condition describes the object category, material, physical properties, and interaction action in natural language when such information is available. The structured attribute condition represents mass, three-axis size, and sliding friction as numerical slots. Each slot stores the attribute value together with indicators of its availability and uncertainty. When an object mesh is available, we additionally compute an optional basis point set (BPS) shape descriptor using 4,096 basis points sampled within a sphere of radius 0.15 m. Table A1 presents representative text conditions and their corresponding structured attributes. The text descriptions are shortened for readability, while the numerical slots retain the physical quantities used by the model. All three context representations are treated as optional during training. We apply attribute dropout, attribute fuzzification, and BPS dropout to expose the model to both accurate simulation metadata and incomplete or uncertain conditions. This training strategy improves robustness in the real-world adaptation setting described in Appendix E, where precise object geometry and physical properties may not be available.

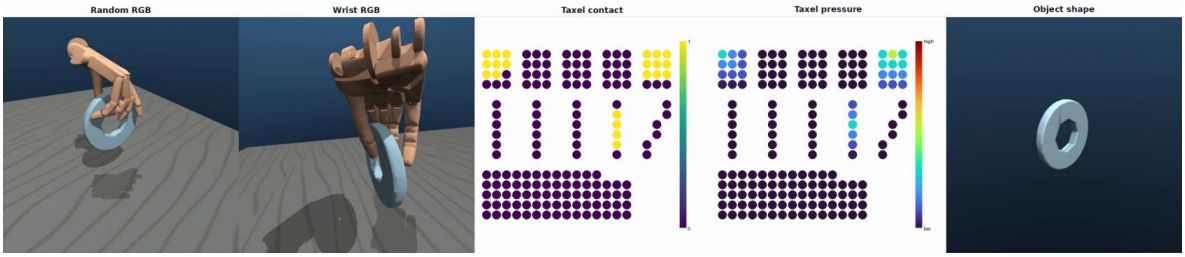
B.2. Dataset Preparation

Manifest Construction. Following simulation, candidate episodes are screened for physical validity and data completeness according to the criteria described in Appendix B.1. From 583,305 candidate episodes, we retain 45,124 complete episodes with synchronized visual observations and interaction-state annotations. The retained episodes are divided into 40,612 training episodes, 2,256 validation episodes, and 2,256 test episodes. To prevent information leakage, we perform the partition at the object-instance level before sampling camera views or temporal

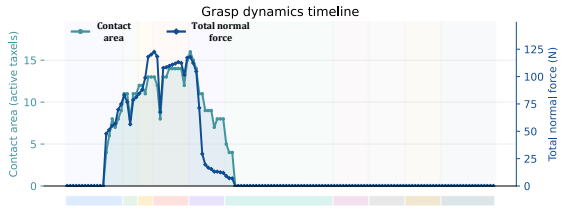


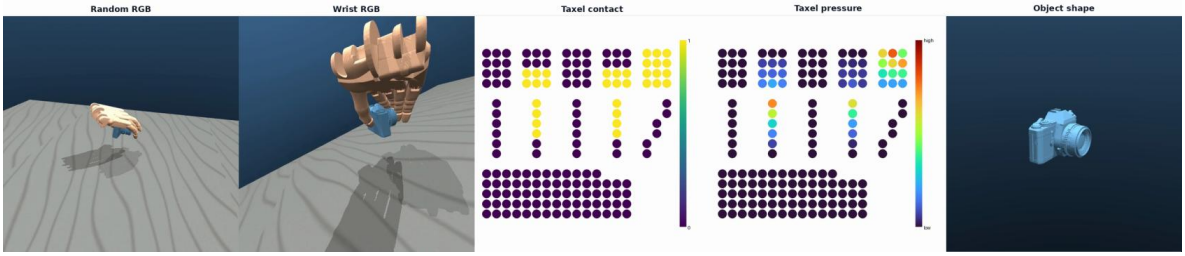
training_text_prompt:
This grasped object is a compact, egg-shaped object. The hand approaches the object from above and grasps it. The object weighs 0.390 kilograms and measures approximately 0.083 by 0.083 by 0.109 meters. It is made of wood, with a sliding friction coefficient of 0.817.
Structured attributes:
category=others, mass=0.3898, size_x=0.0829, size_y=0.0831, size_z=0.1088, material=wood, friction=0.817.





training_text_prompt:
This grasped object is a thin, circular roll of tape. The hand approaches the object from above and grasps it. The object weighs 0.257 kilograms and measures approximately 0.020 by 0.143 by 0.143 meters. It is primarily made of plastic, with a sliding friction coefficient of 0.275.
Structured attributes:
category=others, mass=0.2567, size_x=0.0202, size_y=0.1426, size_z=0.1426, material=plastic, friction=0.275.





training_text_prompt:
This grasped object is a compact, box-shaped camera. The hand approaches the object from above and grasps it. The object weighs 0.139 kilograms and measures approximately 0.056 by 0.059 by 0.091 meters. Its outer casing is primarily made of plastic, with a sliding friction coefficient of 0.646.
Structured attributes:
category=electronics, mass=0.1385, size_x=0.0561, size_y=0.0589, size_z=0.0907, material=plastic, friction=0.646.

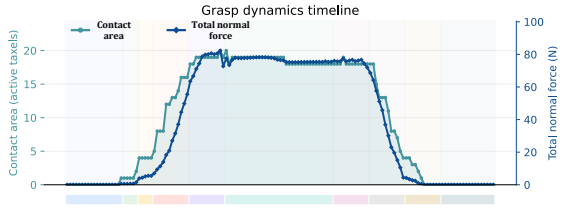


Figure A1. Representative samples from UNISON-Sim. Each sample consists of two rows. The upper row presents synchronized multi-modal observations and interaction states, including random view and wrist-view RGB videos, taxel-level contact and pressure states on the shared hand-surface topology, and object appearances. The lower row shows the corresponding context, including text descriptions and structured attributes, together with the temporal evolution of contact and pressure throughout different grasping stages.

Table A1. Representative object-context entries stored with UNISON-Sim episodes. Text conditions are shortened from the cache field for readability. Structured slots follow the order mass, size along three axes, and sliding friction.

Category	Text condition excerpt	Structured attribute slots
Bottle	An elongated plastic bottle with object mass 0.561 kg and approximate size $0.074 \times 0.074 \times 0.174$ m.	mass 0.5609, size 0.0735/0.0735/0.1738, friction 0.4583
Can	A moderately elongated metal can with object mass 0.237 kg and approximate size $0.058 \times 0.058 \times 0.089$ m.	mass 0.2374, size 0.0584/0.0584/0.0889, friction 0.1861
Bowl	A compact plastic bowl with object mass 0.097 kg and approximate size $0.057 \times 0.075 \times 0.075$ m.	mass 0.0972, size 0.0574/0.0753/0.0753, friction 0.4583
Jar	A compact plastic jar with object mass 0.104 kg and approximate size $0.067 \times 0.071 \times 0.071$ m.	mass 0.1040, size 0.0673/0.0708/0.0708, friction 0.4583
Electronics	A tall and slender metal electronics object with object mass 0.105 kg and approximate size $0.021 \times 0.059 \times 0.105$ m.	mass 0.1054, size 0.0211/0.0589/0.1045, friction 0.1861
Mug	A moderately elongated glass mug with object mass 0.153 kg and approximate size $0.081 \times 0.082 \times 0.110$ m.	mass 0.1534, size 0.0813/0.0819/0.1103, friction 0.2058

clips. All episodes generated from the same object, including different scales, grasp poses, trajectories, viewpoints, and temporal segments, are assigned to the same split. We further verify that each object identifier corresponds to a unique mesh, preventing duplicated geometry from appearing across splits under different identifiers. Consequently, all object instances in the validation and test sets are unseen during training, enabling a stricter evaluation of generalization to novel objects. Category-level split statistics are reported in Table A2.

Table A2. Category-level statistics for the UNISON-Sim partition.

	Object instances			Episodes		
Category	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
Bottle	216	12	12	5,718	318	318
Bowl	128	7	7	1,941	108	108
Box	179	10	10	3,756	209	209
Can	174	10	10	3,961	220	220
Cube	15	1	1	296	15	17
Electronics	184	10	10	3,045	169	169
Food	88	5	5	1,586	88	88
Jar	215	12	12	4,263	237	237
Mug	202	11	11	3,532	196	196
Others	464	26	26	8,399	468	466
Soft	72	4	4	885	49	49
Sphere	36	2	2	774	43	43
Tool	62	3	3	841	47	47
Toy	45	2	2	710	39	39
Vase	42	2	2	905	50	50
All categories	**2,122**	**117**	**117**	**40,612**	**2,256**	**2,256**

Unified State Construction. For each retained episode, the frame-level simulator outputs are converted into the unified state representation. Each RGB frame is aligned with its corresponding CAH pose and the contact and pressure fields defined on the shared 162-taxel topology. Pressure values are processed using the logarithmic force transformation described in Appendix B.1. All normalization statistics for pose, pressure, and structured attributes are computed exclusively from the training split and then fixed for validation and testing.

Temporal Clip Sampling. Training samples are obtained by sampling synchronized temporal clips from the retained episodes. Let T denote the clip length, the default configuration uses $T = 16$. Videos are exported at 15 fps and sampled at 3 fps with a frame stride of 5. The same frame indices are applied to RGB observations, CAH poses, contact fields, pressure fields, and interaction-phase labels, while the episode-level object context is attached to the resulting clip.

C. More Details on Unified State Representation

This appendix section complements the main paper by describing how the unified state representation is organized before it enters the generator or the evaluation protocol.

C.1. Compact Articulated Hand State

The CAH pose state $\mathbf{p}_t$ is designed to make the pose component usable across hands with different morphologies. In a grasping interaction, the state should describe how the palm and fingers move to form the grasp, without requiring every episode to share the same mesh, skeleton, or bone length convention. Direct 3D joints are convenient for scoring. They also tie the target to a particular hand geometry and make retargeting between dexterous hands and human-hand annotations less clean. CAH therefore stores root motion and canonical finger articulation in a 34-dimensional vector. The geometry of a specific hand enters only during visualization or metric computation through forward kinematics. The resulting vector remains compact and fixed across all episodes. It contains three root translation parameters, a 6D root rotation representation, and 25 finger

Algorithm A1 Forward kinematics decoding.

Require: CAH pose state $\mathbf{p}_t$ and its episode-specific MJCF hand model $\mathcal{M}$
Ensure: Decoded 3D hand keypoints $\mathbf{J}_t \in \mathbb{R}^{21\times 3}$
1: Reset $\mathcal{M}$ to its neutral hand configuration
2: Write the enabled finger articulation entries in $\mathbf{p}_t$ to the corresponding hinge coordinates of $\mathcal{M}$
3: Run MuJoCo forward kinematics to obtain the transforms of the palm, finger joints, and fingertips
4: Express the 21 semantic keypoints as palm-frame offsets
5: Decode the 6D root rotation in $\mathbf{p}_t$ through orthogonal completion
6: Transform the palm-frame keypoint offsets into world coordinates using the decoded root rotation and root translation
7: **return** $\mathbf{J}_t$

articulation parameters arranged as five canonical slices for the thumb, index, middle, ring, and little fingers. Each slice represents base flexion, base abduction, and phalanx bending in a shared order. For an embodiment whose kinematic tree lacks one canonical distal degree of freedom, the corresponding slot is kept as a neutral value. This convention preserves a single 34 dimensional state space while allowing the simulator specific mapping to be handled outside the generator.

For evaluation, CAH is converted to the OpenPose [4] hand order with 21 keypoints through the episode MJCF kinematic tree. The conversion is deterministic and uses only kinematic transforms, so pose metrics are not affected by collision resolution or contact constraints in the simulated scene. Algorithm A1 summarizes the deterministic forward-kinematics decoding procedure. During simulation export, the recorded hand state is also decoded once into this 21-keypoint representation. This precomputed skeleton serves as the geometric target for simulated validation episodes, so the target and prediction are evaluated under the same kinematic convention. If the precomputed target is unavailable, the evaluator recomputes it from the recorded CAH sequence using the same steps above. MPJPE, PA-MPJPE, and contact-frame fingertip MPJPE are then computed on the decoded 3D keypoints in mm.

D. Network Architecture Details

This section gives the structures of network modules whose roles are introduced in the main paper, with their architectural details provided in this supplement. We include the condition-side modules that prepare visual, textual, physical-attribute, and shape cues before they enter the State DiT. We also document the pose state encoder and decoder, and provide the taxel connectivity construction used by the tactile topology layer.

D.1. Condition-Side Architectures

Visual LoRA Adapter. The visual backbone is kept frozen, and adaptation is applied only through low-rank residual branches inside selected attention projections. For an input activation $\mathbf{h}$, a frozen linear projection with weight W is augmented as

$$\phi_{\mathrm{LoRA}}(\mathbf{h}) = W\mathbf{h} + \frac{\alpha}{r} BA\mathbf{h}. \tag{7}$$

Here, ϕ_{LoRA} denotes the adapted projection, A and B are trainable low-rank matrices, r is the LoRA rank, and α is the scaling coefficient. This module changes the visual features seen by UNISON while preserving the pretrained backbone projection.

Visual Token Resampler. After DINOv2 [26] produces patch tokens for each video frame, UNISON uses a learned query resampler to obtain a fixed-size set of frame tokens. Let $\mathbf{U}_t$ denote the projected patch-token matrix for frame t, and let $\mathbf{Q}$ denote the learned query matrix shared across frames. The resampler applies query-to-patch multi-head attention followed by a residual feed-forward update,

$$\mathbf{R}_t = \mathbf{Q} + \mathrm{MHA}\left(\mathrm{LN}(\mathbf{Q}), \mathrm{LN}(\mathbf{U}_t), \mathrm{LN}(\mathbf{U}_t)\right), \tag{8}$$

$$\mathbf{V}_t = \mathbf{R}_t + \mathrm{FFN}\left(\mathrm{LN}(\mathbf{R}_t)\right). \tag{9}$$

Here, $\mathbf{R}_t$ is the intermediate attended query set, MHA is multi-head attention, LN is layer normalization, FFN is a two-layer feed-forward network, and $\mathbf{V}_t$ is the compact visual token set for frame t. A learned frame-time embedding is added to $\mathbf{V}_t$ before all frame tokens are flattened into the visual condition sequence.

Structured Attribute Slot Encoder. The structured attribute encoder takes an attribute matrix with one row per physical slot. Each row contains four features: attribute value, known flag, fuzzy-value flag, and fuzzy level. For slot j, the four-dimensional slot feature $\mathbf{s}_j$ is normalized and linearly projected, then added to a learned attribute-type embedding $\mathbf{e}_j$,

$$\mathbf{a}_j = \mathrm{Linear}\left(\mathrm{LN}(\mathbf{s}_j)\right) + \mathbf{e}_j. \tag{10}$$

The slot token is refined by a residual MLP,

$$\tilde{\mathbf{a}}_j = \mathbf{a}_j + \mathrm{MLP}\left(\mathrm{LN}(\mathbf{a}_j)\right). \tag{11}$$

The refined slot tokens are averaged and passed through a final projection to form the global attribute token. This architecture keeps the identity of each physical slot explicit while giving the State DiT one compact object-level physical condition.

BPS Condition Encoder. The BPS encoder has a global path and a token path. The global path takes the full BPS distance vector after radius normalization and applies LayerNorm, Linear, GELU, Dropout, Linear, and a final LayerNorm to produce the global shape token. The token path pads the same distance vector when needed, splits it into equal chunks, and applies a small MLP to each chunk before adding a learned BPS token-type embedding. The current UNISON generator uses the global shape token as the active shape condition, while the token path keeps the interface compatible with token-based shape conditioning. If the BPS input is dropped, the encoder replaces the shape signal with a learned null token or a zero token according to the selected null mode.

Visual-text Alignment Block. The alignment block updates visual and text streams before they are passed to the State DiT. It first normalizes both streams. The visual stream then uses visual tokens as queries and text tokens as keys and values. The text stream uses text tokens as queries and visual tokens as keys and values. Both streams receive residual attention updates followed by separate residual feed-forward updates. This structure gives visual tokens access to object and action semantics, and gives text tokens access to the actual hand-object evidence in the clip.

Global Context Builder. The global context builder prepares the vector that modulates State DiT blocks. Its input consists of the global attribute token and the global BPS token. These tokens are concatenated and passed through a small MLP with LayerNorm, Linear, SiLU [7], and Linear layers, yielding an object-context vector in the shared hidden dimension. The flow-time embedding is then added to this vector to produce the conditioning signal used for AdaLN modulation and attention gates.

D.2. Pose Encoding and Decoding Simplification

An earlier UNISON implementation used a structured pose-token pathway to mirror the tactile token design. It encoded the 34-dimensional CAH pose as seven tokens. One global token received the full CAH vector, one root token received the root translation and 6D rotation slice, and five finger tokens received the thumb, index, middle, ring, and little finger slices. The decoder then predicted the root and finger velocity chunks separately. Each finger chunk used its own finger token together with the global and root tokens, so the local finger prediction could condition on whole-hand and wrist-frame information. This design was anatomically intuitive, but its empirical behavior was worse. The extra pose tokens allowed root and finger states to follow separate hidden trajectories before they were recombined into one CAH vector. In generated results, the model often produced plausible local joint motion while the global hand location became less stable across frames. Small inconsistencies between root and finger token predictions were amplified after forward kinematics, where a single grasp sequence could show visible spatial drift. This failure mode suggests that pose stability depends more on coherent global CAH dynamics than on a high-resolution part topology.

The final UNISON implementation therefore uses a single-token pose path. At each timestep, a linear layer projects the full 34-dimensional CAH vector into one pose token. After State DiT reasoning, the corresponding hidden token is decoded by the temporal final layer, which applies conditioned normalization and a linear prediction head to output the 34-dimensional pose velocity. This simple pathway was unexpectedly strong. It reduced the pose decoder's degrees of freedom, forced all CAH dimensions to share one temporal state before velocity prediction, and preserved global pose coherence more consistently.

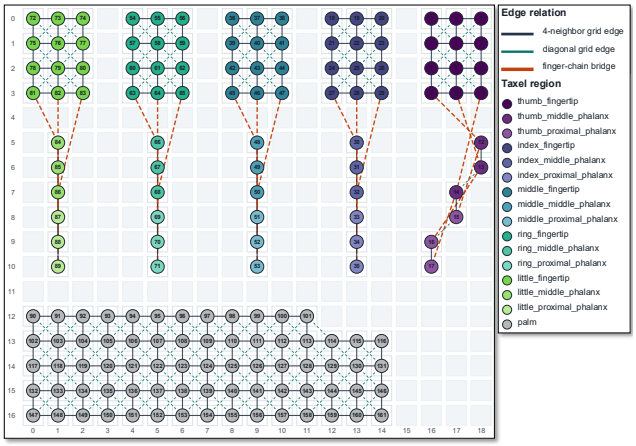


Figure A2. Taxel connectivity graph used by the tactile topology layer. Solid edges encode 4-neighbor grid adjacency, dashed teal edges encode diagonal grid adjacency, and orange bridge edges connect taxel groups along each finger chain. Node colors indicate semantic hand-surface regions used by the region embedding and 16-region pooling.

D.3. Taxel Connectivity Graph

The tactile topology layer uses a fixed connectivity graph over the 162 hand-surface taxels. We denote this graph as $\mathcal{G}_{\text{tax}} = (\mathcal{V}_{\text{tax}}, \mathcal{E}_{\text{tax}})$, where $\mathcal{V}_{\text{tax}}$ is the set of taxels and $\mathcal{E}_{\text{tax}}$ is the set of taxel relations. As illustrated in Figure A2, the graph combines local spatial adjacency within each hand region with anatomical connections along the finger chains. The same graph is shared by the contact and pressure branches and is used only within the tactile encoder and decoder.

The edge set contains three types of relations, which are distinguished by their visual styles in Figure A2. First,

4-neighbor grid edges connect horizontally and vertically adjacent taxels within the two-dimensional layout of each hand region and capture direct local tactile continuity. Second, diagonal grid edges connect oblique neighbors in the same layout, enabling local contact and pressure patterns to propagate across a broader neighborhood. Third, finger-chain bridge edges connect spatially separated taxel groups along the same finger, including the fingertip, middle-phalanx, and proximal-phalanx regions. These bridge relations preserve anatomical continuity that is not fully represented by grid adjacency in the flattened layout. Self-loops are added before graph propagation. Let $\mathbf{A} \in \{0,1\}^{162\times162}$ be the binary adjacency matrix induced by $\mathcal{E}_{\text{tax}}$, and let $\mathbf{D}$ be the diagonal degree matrix of $\mathbf{A}+\mathbf{I}$. The normalized adjacency used by the topology layer is

$$\widetilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}(\mathbf{A}+\mathbf{I})\mathbf{D}^{-\frac{1}{2}},$$

where $\mathbf{I}$ is the identity matrix. This normalization keeps feature magnitudes stable across taxels with different numbers of neighbors. For a taxel feature matrix $\mathbf{U}^{(\ell)} \in \mathbb{R}^{162\times d}$ at layer ℓ, the topology layer performs normalized graph aggregation followed by a residual MLP update:

$$\overline{\mathbf{U}}^{(\ell)} = \widetilde{\mathbf{A}}\,\mathrm{LN}(\mathbf{U}^{(\ell)}), \mathbf{U}^{(\ell+1)} = \mathbf{U}^{(\ell)} + \mathrm{MLP}_{\text{graph}}(\overline{\mathbf{U}}^{(\ell)}).$$

Here, d is the hidden dimension, $\mathrm{LN}(\cdot)$ denotes layer normalization, and $\mathrm{MLP}_{\text{graph}}$ is a lightweight feed-forward update shared across taxel positions within the layer. The same connectivity is used in the encoder before region pooling and in the decoder before producing dense taxel-wise velocities.

E. Real-World Adaptation

UNISON is trained with UNISON-Sim because simulation provides synchronized RGB videos, CAH pose, taxel-level contact, and taxel-level pressure. This training teaches the State DiT a structured pose-contact-pressure prior, yet the visual input at deployment differs from the rendered input used during training. Real RGB videos introduce changes in hand appearance, background, lighting, camera response, motion blur, and occlusion statistics. A simulation-trained visual encoder can therefore present out-of-distribution visual tokens to the state generator when it is applied directly to real videos. Real-world adaptation of UNISON targets this visual input gap. We avoid directly fine-tuning the full model with a small real labeled set, since dense real pose-contact-pressure states are difficult to collect at scale and a limited supervised update may disturb the coupled state prior learned from UNISON-Sim. We instead keep the State DiT and most other modules frozen and adapt only a real-domain LoRA [12] inside the visual encoder. This design matches the source of the domain shift because the pose-contact-pressure generator remains anchored to simulation style video supervision, while the real visual encoder learns to produce tokens compatible with the frozen generator.

Figure A3 gives an overview of the real-world adaptation setting. UNISON-S2R pairs simulated dexterous-hand videos with real human-hand RGB videos at the interaction level, as shown in Fig. A3(a). This paired visual correspondence exposes the domain gap caused by hand embodiment, rendering style, background, camera response, lighting, and occlusion, while avoiding the need for real pose, contact, and pressure labels during adapter training. The adaptation pipeline in Fig. A3(b) restricts this transfer to the visual condition space of UNISON. The simulated branch serves as a frozen teacher with the simulation-domain LoRA, and the real branch serves as a student in which only the real-domain LoRA is updated. Both branches feed their visual tokens into the same frozen State DiT under shared auxiliary conditions. The adapter is therefore trained to make real visual tokens compatible with the frozen state generator, using visual-token alignment, generator-level velocity consistency, and a regularization term that keeps the real-domain LoRA close to the simulation-domain initialization.

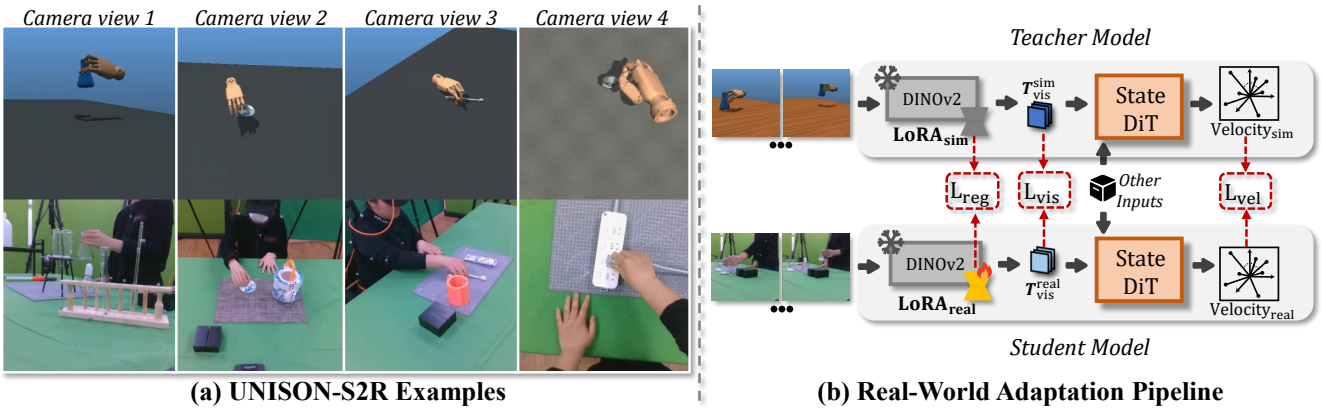
We evaluate the adapted model on a separate EgoTactile-style [48] bare-hand validation set. The set provides synchronized contact and pressure measurements for real RGB videos and lacks hand-pose ground truth. We therefore study pose through image-aligned MANO [33] retargeting and agreement with a WiLoR [30] pseudo-reference. All validation signals remain outside model training. The following subsections describe the paired data, the visual adapter, and the real validation protocol.

E.1. UNISON-S2R Paired Visual Transfer Data

UNISON-S2R is constructed for visual-domain transfer rather than dense real state supervision. Each paired record contains a real human-hand RGB clip and a corresponding simulated dexterous-hand RGB clip from an aligned action segment. We write the paired dataset as

$$\mathcal{D}_{\text{S2R}} = \left\{ \left(\mathbf{x}^{\text{real}}_{1:T}, \mathbf{x}^{\text{sim}}_{1:T}\right)_i \right\}_{i=1}^{N}, \tag{12}$$

where i indexes the i-th paired record. Each pair also stores metadata for the source sequence, camera view, frame indices, hand-side type, and action primitive. The primitive is an action label such as *cap*, *pour*, *scoop*, or *hold*, and is used to construct the shared action prompt for adapter training. The pair construction uses complementary information from DexManipNet [19] and OakInk2 [49]. DexManipNet provides dexterous-hand manipulation trajectories and action sequence metadata, including the primitive label, hand-side type, and temporal span. OakInk2 provides multi-view real human hand-object RGB demonstrations. For each aligned record, we select the OakInk2 camera stream and time interval associated with the real demonstration, then



(a) UNISON-S2R Examples **(b) Real-World Adaptation Pipeline**

Figure A3. Real-world adaptation with UNISON-S2R paired visual transfer data. (a) UNISON-S2R provides temporally paired simulated dexterous-hand clips and real human-hand RGB clips for the same interaction, giving a visual correspondence across simulation and real domains without requiring dense real state labels for adapter training. (b) The adaptation pipeline follows a teacher-student design. The teacher branch encodes simulated clips with the frozen simulation-domain visual encoder, while the student branch encodes real clips with a trainable real-domain LoRA. The State DiT and state decoders are shared and frozen, so adaptation is confined to the visual condition interface through token alignment, velocity consistency, and LoRA regularization.

Table A3. UNISON-S2R paired visual transfer data. The dataset provides synchronized real/sim RGB correspondence for visual-interface adaptation while excluding real pose-contact-pressure labels from adapter training.

Item	Value
Paired records	4,820
Source demonstrations	488
Real & simulated frames	3.74M
Average sequence length	388 frames
Camera views	4 synchronized views
Action primitive labels	61, e.g., cap, pour, hold
Hand-side coverage	right, left, bimanual
Frame rate and resolution	60 fps, 848×480

render a MuJoCo dexterous-hand video for the matched action segment and camera view. The result is a pair of videos with the same frame count, frame rate, and viewpoint index, which gives the adapter a real/sim visual correspondence without requiring real pose, contact, or pressure labels.

Table A3 summarizes the UNISON-S2R manifest. It contains 4,820 paired sequence-view records from 1,205 aligned sequence IDs and 488 source demonstrations. The four camera views cover synchronized viewpoints of the same interaction, and the 61 primitive labels cover diverse manipulation actions such as capping, pouring, scooping, holding, and placing. Across real and simulated domains, the manifest contains approximately 3.74 million RGB frames at 60 fps and 848×480 resolution. UNISON-S2R intentionally keeps object-side conditions weak. The paired records do not provide reliable real object mass, size, friction, material, or mesh geometry. The teacher and student branches therefore receive the same action-level text, null structured attributes, and no BPS shape input. The default prompt is *"A hand performs {primitive} with an object."*, and the generic fallback is *"A hand is manipulating an object."* This protocol focuses training on the RGB domain gap and prevents the adapter from using unverified real object conditions.

Beyond its use in UNISON, UNISON-S2R provides a reusable real/sim video correspondence for dexterous manipulation. To our knowledge, it is one of the first paired real/sim RGB resources built for unified hand-state adaptation in hand-object interaction. The same correspondence can support studies on visual domain adaptation, real-to-sim alignment, robot-hand retargeting, and action-conditioned simulation for manipulation. This makes the dataset a transfer resource rather than only an internal component of the UNISON training pipeline.

E.2. Frozen-UNISON Visual Adapter

Architecture. The adapter uses a teacher and student structure after UNISON has been trained on UNISON-Sim. The teacher is the simulation-trained UNISON model, including the DINOv2 [26] visual encoder with simulation LoRA [12], the text and object-condition encoders, the State DiT, and the state decoders. All teacher parameters are frozen. The student starts from the same checkpoint, while only the real-domain visual LoRA parameters inside the visual encoder are trainable. The real-domain LoRA is initialized from the simulation LoRA, so adaptation starts from a visual representation already useful for hand-object state generation.

Shared Conditions. For a paired sample, the teacher branch encodes $\mathbf{x}_{1:T}^{\text{sim}}$ with the frozen simulation visual encoder, while the student branch encodes $\mathbf{x}_{1:T}^{\text{real}}$ with the trainable real-domain LoRA inside the visual encoder. Both branches use the same text prompt, the same random flow state $\mathbf{z}_\tau$, the same flow time τ, null structured attributes, and no BPS shape input:

$$\mathbf{o}_{\text{text}}^{\text{real}} = \mathbf{o}_{\text{text}}^{\text{sim}}, \mathbf{o}_{\text{attr}}^{\text{real}} = \mathbf{o}_{\text{attr}}^{\text{sim}} = \emptyset, \mathbf{o}_{\text{shape}}^{\text{real}} = \mathbf{o}_{\text{shape}}^{\text{sim}} = \emptyset. \tag{13}$$

This shared-condition protocol ensures that the adapter objective is driven by the visual domain gap rather than by differences in text, object attributes, or shape priors. It also matches the weak-condition curriculum, where the simulation-trained generator is exposed to missing or imprecise object context. Let E_{sim} denote the frozen simulation visual encoder, E_{real} denote the real-domain visual encoder with trainable LoRA, and G denote the frozen UNISON state generator, including the State DiT and state decoders. We collect the shared text prompt together with the null attribute and shape entries in Eq. (13) as $\mathbf{o}$. Under this shared condition, the two branches differ only in their visual tokens. Therefore, feeding the simulated and real visual tokens into the same frozen generator yields a teacher velocity and a student velocity that can be directly aligned:

$$\begin{aligned} \mathbf{v}^{\text{sim}} &= G\left(\mathbf{z}_\tau, \tau, E_{\text{sim}}(\mathbf{x}_{1:T}^{\text{sim}}), \mathbf{o}\right), \\ \mathbf{v}^{\text{real}} &= G\left(\mathbf{z}_\tau, \tau, E_{\text{real}}(\mathbf{x}_{1:T}^{\text{real}}), \mathbf{o}\right). \end{aligned} \tag{14}$$

Here, $\mathbf{v}^{\text{sim}}$ serves as the teacher state-space flow velocities predicted by the simulation branch, while $\mathbf{v}^{\text{real}}$ is the student prediction from the real-domain branch. During adaptation, E_{sim} and G are kept frozen, and only the LoRA parameters inside E_{real} are updated.

Objective. The training objective aligns the real visual encoder with the frozen generator interface. The visual feature loss compares the frame-level visual tokens consumed by the State DiT:

$$\mathcal{L}_{\text{vis}} = \mathbb{E}\left[1 - \cos\left(\mathbf{T}_{\text{vis}}^{\text{real}}, \text{sg}(\mathbf{T}_{\text{vis}}^{\text{sim}})\right)\right], \tag{15}$$

where $\mathbf{T}_{\text{vis}}^{\text{real}}$ and $\mathbf{T}_{\text{vis}}^{\text{sim}}$ denote the frame-level visual tokens from the real and simulated branches, and $\text{sg}(\cdot)$ denotes stop-gradient. Visual feature alignment is paired with velocity consistency loss through the frozen generator:

$$\mathcal{L}_{\text{vel}} = \left\|\mathbf{v}^{\text{real}} - \text{sg}(\mathbf{v}^{\text{sim}})\right\|_2^2. \tag{16}$$

This term checks whether the real visual tokens lead the frozen generator toward the same state-space velocity as the paired simulated clip. The LoRA regularizer keeps the real adapter near the simulation initialization:

$$\mathcal{L}_{\text{reg}} = \frac{1}{|\Theta_{\text{LoRA}}|} \sum_{\theta_i \in \Theta_{\text{LoRA}}} \left\|\theta_i^{\text{real}} - \text{sg}(\theta_i^{\text{sim}})\right\|_2^2. \tag{17}$$

Here, Θ_{LoRA} is the set of trainable LoRA parameters inside E_{real}, and $|\Theta_{\text{LoRA}}|$ is the number of such parameters. The full objective is

$$\mathcal{L}_{\text{S2R}} = \lambda_{\text{vis}}\mathcal{L}_{\text{vis}} + w_{\text{vel}}(s)\lambda_{\text{vel}}\mathcal{L}_{\text{vel}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \tag{18}$$

where s denotes the optimization step, $w_{\text{vel}}(s)$ warms up the velocity loss during early training, and the λ terms are scalar loss weights. In our implementation, we set $\lambda_{\text{vis}} = 1.0$, $\lambda_{\text{vel}} = 1.0$, and $\lambda_{\text{reg}} = 10^{-4}$.

E.3. Bare Hand Validation

Evaluation Setting. We collected a separate EgoTactile-style [48] bare-hand validation set covering 13 everyday objects. It comprises 75 video clips with 16,406 RGB frames in total, corresponding to 18.23 minutes of recording. The videos were captured at a resolution of 1280×720 and 15 FPS, with three to nine clips collected for each object. Before recording, we measured the weight and dimensions of each object and assigned an approximate friction coefficient based on its surface material. Each clip is accompanied by synchronized taxel-level contact and pressure measurements. The entire validation set, including all video clips and associated physical measurements, is held out from training and used exclusively for evaluation. Because motion-capture and MANO pose annotations are unavailable, contact and pressure are evaluated directly against the synchronized taxel measurements, whereas pose is assessed using the agreement protocol described below.

Bare-Hand Acquisition Protocol. We follow the bare-hand setting of EgoTactile. The operator wears the tactile glove on the dominant right hand, while the bare left hand performs the matched grasping motion recorded by the RGB camera. The camera observes only the bare left hand and the manipulated object. A metronome guides both hands to follow the same grasping rhythm and provides temporal correspondence between the left-hand video and the right-hand tactile sequence. The glove contains 162 taxels whose anatomical distribution exactly matches the tactile topology used by UNISON. The synchronized measurements therefore follow a direct taxel-to-taxel correspondence without spatial interpolation. Figure A4 summarizes the bare-hand data acquisition protocol, including the physical taxel layout of the tactile glove, its mapping onto the MANO hand surface, and the corresponding data collection setup. Both representations contain 162 taxels, providing a shared topology for aligning tactile measurements with the reconstructed hand geometry.

Pressure Scale Calibration. The pressure channel requires an additional scale calibration before simulated predictions can be compared with real tactile-glove measurements. The simulator reports taxel-level normal pressure,

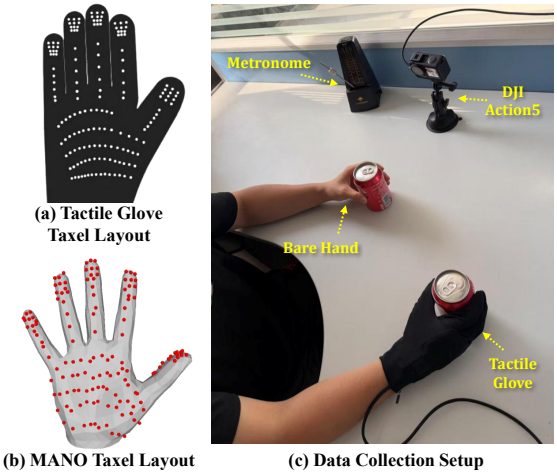


Figure A4. Overview of the bare-hand data acquisition protocol. (a) The tactile glove contains 162 taxels distributed across the fingers and palm. (b) The same 162 taxel locations are mapped one-to-one onto the MANO hand surface, establishing a shared topology between physical tactile measurements and reconstructed hand geometry. (c) The data collection setup, in which the bare and gloved hands manipulate matched objects while the camera records the interaction and the metronome provides timing cues.

while the glove reports sensor-dependent tactile readings. These two measurements are aligned by taxel location and time, yet their numerical scales and low-response behavior differ. Directly using the simulated magnitude as a real pressure value would therefore confound model error with a sensor-scale mismatch. We calibrate only the scalar pressure magnitude and keep the taxel topology, contact labels, and temporal alignment unchanged. The real-domain calibration statistics are estimated from 600 grasp records in the EgoTactile dataset. EgoTactile uses the same 162-taxel tactile glove and the same bare-hand acquisition setting as our validation protocol. These external calibration records are separate from the 75 evaluation clips. The calibration parameters are fixed before final evaluation, and no evaluation clip contributes to the fitted pressure scale. We first estimate a contact threshold separately for the simulated and real pressure distributions. For each domain, we collect positive taxel pressure values and take the 95th percentile of the lower half of this distribution as the weak-contact cutoff. This rule removes background glove readings and near-zero simulated contact responses using the same robust statistic in both domains. It gives thresholds of $\delta_{\text{sim}} = 2.62$ N for simulated pressure and $\delta_{\text{real}} = 14$ N for real glove pressure. Table A4 reports the pressure statistics before and after thresholding.

Let $\mathcal{S}$ and $\mathcal{R}$ denote the simulated and real pressure value sets after applying the thresholds above. Here each element

Table A4. Pressure statistics used for simulation-to-real pressure-scale calibration. The real statistics are computed from 600 external EgoTactile grasp records collected with the same 162-taxel glove. The contact rows are used to fit the calibration map, while the nonzero rows show the unfiltered scale of each domain. Filter thresholds and all pressure statistics are expressed in N.

Source	Records	Filter	Mean	Median	P95	P99	Max
Simulation nonzero	5,000	> 0	3.51	2.77	8.70	13.25	158.12
Simulation contact	5,000	> 2.62	5.42	4.68	10.32	15.70	158.12
EgoTactile nonzero	600	> 0	21.16	14.00	65.00	94.00	255.00
EgoTactile contact	600	> 14	36.77	31.00	78.00	108.00	255.00

is a scalar taxel pressure magnitude. We model the calibration as a monotonic power-law map because it covers both a global scale shift and a mild nonlinear sensor response with only two parameters. Let $Q_{\mathcal{S}}(q)$ and $Q_{\mathcal{R}}(q)$ be the empirical q-quantiles of $\mathcal{S}$ and $\mathcal{R}$. We estimate the calibration scale a and exponent b by matching quantiles in log space,

$$(a, b) = \arg\min_{a,b} \sum_{q \in \mathcal{Q}} \left[\log Q_{\mathcal{R}}(q) - \log a - b \log Q_{\mathcal{S}}(q)\right]^2, \tag{19}$$

where $\mathcal{Q} = \{0.10, 0.20, \ldots, 0.90, 0.95\}$. The fitted parameters are $a = 4.65$ and $b = 1.22$, with a mean relative quantile error of 1.93%. The calibrated real-scale pressure for a simulated scalar pressure ρ^{sim} is then

$$\hat{\rho}^{\text{real}} = \begin{cases} 0, & \rho^{\text{sim}} \le \delta_{\text{sim}}, \\ \min\left(4.65(\rho^{\text{sim}})^{1.22}, 108\right), & \rho^{\text{sim}} > \delta_{\text{sim}}. \end{cases} \tag{20}$$

The upper bound 108 N is the 99th percentile of the real contact-pressure distribution. It limits rare simulated outliers while preserving the observed real pressure range used by the evaluation metrics.

MANO Retargeting from CAH Pose. UNISON is trained on a large collection of Shadow Hand interactions and can recover plausible grasp articulation even when the hand is heavily occluded by the manipulated object. However, the Shadow Hand and the human hand differ substantially in bone lengths and kinematic proportions. Directly comparing their joint coordinates would therefore mix pose quality with embodiment-specific geometry. To enable human-hand visualization and pose analysis, we retarget the generated CAH pose to MANO. The retargeting preserves the palm orientation and relative finger articulation predicted by UNISON while adapting them to human-hand geometry. For each frame, the generated CAH pose is first decoded by Shadow Hand forward kinematics into a common 21-joint semantic skeleton. We then optimize a MANO pose whose bone directions follow this decoded skeleton. WiLoR provides the MANO shape, camera pa-

rameters, and image-space 2D joints used during retargeting. Its estimated 3D pose is excluded from both initialization and supervision. Consequently, the articulation signal is derived from the UNISON prediction, while WiLoR contributes only the human-hand shape and image registration cues. The MANO shape and camera parameters are fixed during optimization. The MANO pose is obtained by minimizing

$$\mathcal{L}_{\text{ret}} = \lambda_{\text{dir}}\mathcal{L}_{\text{dir}} + \lambda_{\text{proj}}\mathcal{L}_{\text{proj}}, \tag{21}$$

where $\mathcal{L}_{\text{dir}}$ measures the mean cosine distance between corresponding unit bone directions of the MANO and Shadow Hand skeletons. Because this term depends on bone directions rather than joint coordinates, it transfers finger articulation without imposing Shadow Hand bone lengths. The projection loss $\mathcal{L}_{\text{proj}}$ measures the mean absolute distance between the projected MANO joints and the WiLoR 2D joints, with image coordinates normalized by the image width and height. This term keeps the retargeted hand aligned with the observed RGB frame. Both losses are computed on the same 21-joint semantic skeleton. We use $\lambda_{\text{dir}} = 1.0$ and $\lambda_{\text{proj}} = 1$ for all clips. The first frame is initialized from the MANO mean pose, and each subsequent frame is initialized from the retargeted pose of the preceding frame to improve temporal stability. We optimize the MANO pose for 100 iterations using Adam with a learning rate of 10^{-2}. The optimized MANO parameters are then used to generate the hand mesh for rendering and pose analysis. This retargeting is applied only as post-processing and does not modify or fine-tune the UNISON generator. Figure 8 in the main paper and additional Figs. A5–A7 here illustrate the resulting image-aligned MANO hands.

Pose Evaluation with Pseudo-Reference. The tactile glove we use does not provide calibrated 3D hand joints, so the bare-hand set has no pose ground truth. We use the WiLoR reconstruction as a pseudo-reference in Table 4. The retargeted UNISON pose and the WiLoR reconstruction are converted to a common 21-joint semantic skeleton before computing MPJPE, PA-MPJPE, and contact-frame fingertip MPJPE. We use Δ to denote the difference from the fixed WiLoR pseudo-reference. WiLoR provides the MANO shape, camera parameters, and 2D registration cues during retargeting, while its estimated 3D pose is excluded from initialization and supervision. The reported pose values therefore provide a diagnostic measure of the reconstructed 3D pose relative to the WiLoR reference.

Results on Real Videos. Table 4 reports a diagnostic ΔMPJPE of 10.5 mm, 3.5 mm ΔPA-MPJPE, and 14.1 mm ΔCF-Ftip between the retargeted UNISON pose and the fixed WiLoR pseudo-reference. The lower ΔPA-MPJPE results from removing the global similarity transform, whereas ΔCF-Ftip is more sensitive to local articulation differences at the fingertips. The main-paper visualization in Fig. 8 and additional Figs. A5–A7 consistently show articulated finger configurations that align with the visible hand and remain stable under hand-object occlusion. This broad visual evidence demonstrates that UNISON can accurately estimate 3D grasp poses in challenging in-the-wild scenes after visual-domain adaptation.

In the main-paper Fig. 8 and additional Fig. A5, the grasped objects occlude most of the thumb. Under this visual ambiguity, the WiLoR reconstruction exhibits an overly closed thumb configuration that is difficult to reconcile with the observed grasp. UNISON produces a wider and more plausible thumb articulation that better agrees with the current hand-object configuration. Figs. A6 and A7 present more severe occlusion, where only a small portion of the hand remains visible. Although WiLoR fits the locally visible hand region, its completed pose can be inconsistent with a physically plausible grasp. UNISON recovers a more coherent grasp configuration by leveraging the learned temporal and interaction-state priors. When the hand is partially truncated by the image boundary, the WiLoR reconstructions also exhibit small pose shifts and temporal jitter, whereas the UNISON trajectories remain more stable across frames. These observations highlight the advantage of interaction-aware temporal modeling when direct visual evidence is incomplete or ambiguous. The generated contact and pressure distributions in the main-paper Fig. 8 and additional Figs. A5–A7 also agree closely with the ground truth. Across the displayed interactions, UNISON consistently recovers the principal contact regions and pressure-bearing locations, including under substantial hand-object occlusion. The identical pressure values observed over the six proximal and middle-finger taxels in the ground truth are caused by the tactile glove's hardware-level signal processing. To suppress pressure artifacts introduced by glove folding during grasping, the glove averages the measurements from these six taxels and assigns the averaged value to each location. Despite this hardware-induced spatial smoothing, UNISON captures the dominant pressure patterns and their temporal evolution. Together, the pose, contact, and pressure results show that visual-domain adaptation enables UNISON to generate physically coherent and temporally stable hand interaction states in challenging in-the-wild videos.

F. Additional Experimental Details

This section provides the training and evaluation details that support the experiments in the main paper. We first specify the shared experimental configuration and clarify the criteria used to select baseline methods. We then describe how prior methods are adapted and converted to the common UNISON state space for comparison. Finally, we present

the unified cross-embodiment pose evaluation protocol and the implementation details of the pressure-related metrics. Together, these specifications establish consistent data, output, and evaluation conventions across all compared methods.

F.1. Training and Evaluation Configuration

Unless an experiment explicitly changes the category split, UNISON-Sim experiments use the train, validation, and test partitions described in Appendix B.2. Training samples are length-T clips with $T = 16$, and all state targets are sampled at the same frame indices as the RGB observations. The default visual input resolution is 480×480, and the default test view is the random oblique stream. The split is resolved before constructing clip-level samples. During training, the loader draws a valid temporal start and one available camera view from training episodes only. For validation and testing, it uses the fixed random view and selects one deterministic temporal window per episode using seed 42. Thus, an episode contributes one reproducible evaluation clip, while alternative views and overlapping windows from that episode cannot enter another partition. The training split alone is used to optimize model parameters and fit all data-normalization statistics. The validation split is used for checkpoint selection according to the lowest total validation loss. The test split is held out from both optimization and checkpoint selection and is evaluated only with the selected checkpoint. UNISON, its ablations, and all retrained baselines use the same partition manifest. The full UNISON model and its ablation variants are trained with the same optimizer family, training schedule, checkpoint selection rule, and evaluation script unless a variant explicitly changes the evaluated component. Baselines that are retrained on UNISON-Sim follow the same train split and are evaluated with the same metric implementation after their outputs are converted to the common pose and tactile layouts. All reported UNISON results use Euler integration with 30 steps. At each step, the full hierarchical model makes three ordered State DiT passes for pose, contact, and pressure, for 90 State DiT passes per sampled trajectory. The parallel state-generation variants use one State DiT pass per step, for 30 passes per trajectory. The number of integration steps is fixed across these variants.

F.2. Baseline Scope and Excluded Methods

The main comparisons include methods whose inference inputs can be aligned with single-view RGB video and optional weak object context. This scope is important because UNISON-Sim evaluates generation of pose, contact, and pressure from visual observations, with object geometry and physical attributes used only when they are part of the benchmark condition. Methods that assume complete hand-object geometry at the contact-estimation stage are therefore outside the main comparison protocol. This criterion excludes several classic contact-oriented methods from the main tables. ContactOpt [8] infers desired contact using hand and object meshes and then optimizes the hand pose. CPF [44] also relies on hand-object pose and geometry estimates before its contact-aware fitting stage. Including either method would require an additional 3D hand-object reconstruction pipeline. This would change the benchmark input assumption and make the comparison less direct.

F.3. Baseline Adaptation Protocol

Most prior methods use input, output, or embodiment representations that differ from those of UNISON. Pose methods commonly predict MANO-based meshes, joints, or trajectories, while contact and pressure methods operate on method-specific hand meshes or image-space representations. These predictions cannot be directly evaluated against the CAH pose representation and the 162-taxel contact-pressure topology. We therefore perform method-specific adaptations to align each baseline with the UNISON-Sim task, data interface, and hand embodiment. The adaptations preserve the core modeling principle of each method while modifying the necessary input modules, prediction heads, and representation-dependent components.

For pose baselines, including HaMeR [28] and HaPTIC [45], the original MANO-dependent components are adapted to predict the Shadow Hand state used in UNISON-Sim. Depending on the method, this adaptation includes changes to the input processing, pose parameterization, prediction head, and a small number of representation-specific network components. The adapted models are then trained or fine-tuned on UNISON-Sim training split. Their predicted states are converted to the common semantic 3D joint set before metric computation, following the unified pose evaluation protocol described in Appendix F.5. For contact prediction, ContactPrompt [15] retains its original training-free reasoning framework, while its MANO-specific geometric prompts are replaced with Shadow Hand region and taxel prompts. The adapted method identifies contacted hand regions and directly predicts binary contact values on the 162-taxel topology, without UNISON-Sim-specific parameter training. For pressure and tactile baselines, including EgoPressure [54], TouchAnything [56], and EgoTactile [48], the input pipeline and prediction interface are adapted when required by the UNISON-Sim setting. Their native pressure or tactile outputs are converted to the 162-taxel pressure representation. Baselines marked with an asterisk in Table 2 are trained or fine-tuned on UNISON-Sim training split after the required adaptation. All adapted methods use the same training split and the supervision available under the corresponding benchmark setting.

The auxiliary inputs in Table 2 follow the design of each

method rather than being artificially unified. HaMeR and HaPTIC use RGB observations, while ContactPrompt additionally uses hand-region and taxel prompts. EgoPressure and TouchAnything use RGB inputs in our evaluation, whereas EgoTactile also incorporates detailed text descriptions and hand and object masks. UNISON uses episode-level text, structured physical attributes, and an optional BPS shape prior. These conditions are integral to the respective methods, and enforcing identical inputs would either remove key components or require modifications beyond their original designs. In particular, UNISON leverages object context to resolve pressure ambiguities that cannot be determined from RGB alone. To further examine the effect of auxiliary inputs, Table A5 compares UNISON with TimeSformer under identical conditioning. Moreover, the Context Robustness Curriculum in Section 5.3 prepares UNISON for missing object information, while Table A7 evaluates the same model with different auxiliary inputs removed at inference. Even with only fixed generic text, UNISON achieves 0.720 contact F1 and 0.249 pressure vIoU, demonstrating its ability to operate without detailed object context.

F.4. Pressure-to-Contact and Post-processing Protocol

All pressure predictions are first converted to the common 162-taxel layout. For each pressure-output method, we compute the 99th percentile of its positive validation predictions as a method-specific scale. The pressure field is normalized by this frozen scale, and one shared normalized contact threshold selected on the validation split is then applied to every method without an explicit contact output. The scale and threshold are fixed before test evaluation. This validation-calibrated procedure accommodates different native pressure ranges without method-specific test tuning. The same post-processing is used for UNISON and every adapted pressure baseline. It consists of temporal smoothing, non-negative pressure clamping, and pressure-to-contact conversion when required. Methods with an explicit binary contact output retain that output and use the shared smoothing convention. Pressure-derived contact results are reported under the same protocol in all main and ablation tables.

F.5. Unified Cross-Embodiment Pose Protocol

Pose methods use MANO parameters, meshes, direct joints, or the CAH state used by UNISON. Their outputs are converted to a common 21-joint semantic skeleton before metric computation. For UNISON-Sim evaluation, the recorded and predicted CAH states are decoded using the same forward-kinematics convention described in Appendix C.1. The resulting simulation joints provide the targets for MPJPE, PA-MPJPE, and contact-frame fingertip MPJPE in Table 2.

For the real bare-hand evaluation, the dataset reported in Table 4 lacks pose ground truth. WiLoR [30] therefore provides the fixed pseudo-reference skeleton and the registration quantities used by the MANO retargeting procedure described in Appendix E.3. UNISON first decodes its CAH pose through Shadow Hand forward kinematics and then optimizes a MANO pose using the direction and projection terms in Eq. (21). The WiLoR 3D pose is excluded from this optimization. Its shape, camera, and projected joints provide human-hand geometry and image registration cues. We use ΔMPJPE, ΔPA-MPJPE, and ΔCF-Ftip to denote the corresponding pose differences from the fixed WiLoR pseudo-reference. Measured contact annotations determine the frames included in ΔCF-Ftip. These pose values diagnose the reconstructed 3D pose relative to WiLoR, while contact and pressure retain direct evaluation against synchronized measurements.

F.6. Metric Implementation Details

Pressure vIoU is computed on the non-negative normalized pressure field. Let $\hat{r}_{t,i}$ denote the predicted pressure value for taxel i at frame t, and let $r_{t,i}$ denote the corresponding ground-truth pressure value. The volumetric overlap is computed as

$$\mathrm{vIoU} = \frac{\sum_{t=1}^{T}\sum_{i=1}^{162}\min(\hat{r}_{t,i}, r_{t,i})}{\sum_{t=1}^{T}\sum_{i=1}^{162}\max(\hat{r}_{t,i}, r_{t,i})}. \tag{22}$$

Sequences with zero pressure mass in both prediction and ground truth are excluded from this pressure-overlap computation. Part-wise CoP error measures whether pressure mass is localized on the correct anatomical hand region. We use the six EgoTactile parts, namely the five fingertips and the palm. For part g, let $\mathcal{I}_g$ be its taxel indices, and let $\mathbf{u}_i \in \mathbb{R}^2$ be the fixed layout coordinate of taxel i. The predicted center of pressure for part g at frame t is

$$\hat{\mathbf{q}}_{t,g} = \frac{\sum_{i\in\mathcal{I}_g}\hat{r}_{t,i}\mathbf{u}_i}{\sum_{i\in\mathcal{I}_g}\hat{r}_{t,i} + \epsilon},$$

where ϵ is a small numerical constant. The ground-truth center $\mathbf{q}_{t,g}$ is computed in the same way from $r_{t,i}$. Following EgoTactile, Ω_g denotes the valid frames for part g where a frame is included when more than 20% of the taxels in $\mathcal{I}_g$ exceed the pressure threshold τ_{CoP} in either prediction or ground truth. The reported CoP error is the mean localization error over valid frames and parts:

$$\mathrm{CoP} = \mathrm{mean}_g\left[\frac{1}{|\Omega_g|}\sum_{t\in\Omega_g}\|\hat{\mathbf{q}}_{t,g} - \mathbf{q}_{t,g}\|_2\right],$$

where the outer mean is taken over the six parts with at least one valid frame. For methods that output pressure without

an explicit contact map, contact is derived by thresholding the predicted pressure values on the same 162-taxel layout.

G. Additional Experimental Analysis

This section provides additional experimental analyses that complement the main results. We include more ablation studies and supplementary qualitative visualizations. All quantitative results follow the same evaluation protocol as the main experiments. Unless otherwise stated, each table uses the shared UNISON-Sim split, the common pose-contact-pressure metrics, and the output conversion rules described above.

Deterministic Baseline. We construct a deterministic baseline by adapting TimeSformer [2] to the unified state outputs. The backbone processes the same 16-frame clips as UNISON using divided space-time attention and receives the same text, structured-attribute, and BPS conditions. A pose head maps the spatially pooled frame tokens to the 34-dimensional CAH pose. For tactile prediction, 162 learnable taxel queries attend to the visual and condition tokens at each frame. Each query produces a contact logit and logits over nine log-spaced pressure bins [9, 48]. Continuous pressure is recovered by the expectation over bin centers and gated by the predicted contact probability. All three states are predicted in parallel without hierarchical state conditioning. The TimeSformer backbone is initialized from Kinetics-400 [16] pretraining, and its temporal embeddings are interpolated from 8 to 16 frames. All parameters are fine-tuned on UNISON-Sim training split using inputs resized to 480×480. Training uses AdamW for 30 epochs with a batch size of 4 per GPU and weight decay 0.05. The backbone learning rate is 2×10^{-5}, while the condition projections and prediction heads use 2×10^{-4}. The objective contains three normalized terms. Smooth L1 loss supervises pose, focal loss supervises taxel contact, and pressure-bin cross-entropy is computed only on ground-truth contact taxels. Their weights are 1.0, 2.0, and 1.0, respectively. The larger contact weight counteracts the strong sparsity of active taxels. Table A5 shows that TimeSformer* provides a usable pose estimate, reaching 30.8 mm MPJPE compared with 18.4 mm for UNISON. The tactile gap is substantially larger. Contact F1 decreases from 0.756 to 0.574, while pressure vIoU decreases from 0.295 to 0.139 and CoP error increases from 4.19 to 7.36. Pose is strongly constrained by visible hand kinematics, whereas contact and pressure remain sparse and partially observed under hand-object occlusion. A deterministic predictor must map these ambiguous observations to one output, which favors conservative contact activation and weak pressure peaks. UNISON instead models a conditional distribution over complete state trajectories through Flow Matching. Cross-state interaction and hierarchical conditioning allow pose and contact evidence to guide pressure generation, making the generative formulation better suited to recovering tactile states under visual ambiguity.

Table A5. Comparison with a deterministic baseline on UNISON-Sim. Both methods use the same state supervision and auxiliary conditions. TimeSformer* denotes our adaptation of TimeSformer to the unified CAH pose, contact, and pressure outputs.

Method	Pose			Contact			Pressure		
	MPJPE ↓	PA ↓	CF-Ftip ↓	F1 ↑	IoU ↑	T-Acc. ↑	MAE ↓	vIoU ↑	CoP ↓
TimeSformer*	30.8	10.9	37.1	0.574	0.402	0.873	1.47	0.139	7.36
UNISON	**18.4**	**4.5**	**18.7**	**0.756**	**0.607**	**0.947**	**0.46**	**0.295**	**4.19**

Table A6. Architecture and training strategy ablation on UNISON-Sim. Each variant removes or replaces one component of the full UNISON model while keeping the remaining training and evaluation protocol unchanged. "attn." and "attr." denote attention and attributes, respectively.

Variant	Pose			Contact			Pressure		
	MPJPE ↓	PA ↓	CF-Ftip ↓	F1 ↑	IoU ↑	T-Acc. ↑	MAE ↓	vIoU ↑	CoP ↓
UNISON	**18.4**	4.5	**18.7**	**0.756**	**0.607**	**0.947**	**0.46**	**0.295**	**4.19**
w/o taxel topology	18.9	**4.4**	19.4	0.743	0.591	0.941	0.73	0.216	5.18
w/o local visual attn.	**18.2**	4.6	18.8	0.748	0.597	0.943	0.50	0.277	4.26
w/o structured attr.	18.6	4.5	19.1	0.754	0.605	0.944	0.78	0.221	4.49
w/o BPS prior	18.5	**4.4**	18.9	0.749	0.599	0.938	0.60	0.275	4.58
w/o teacher forcing	20.9	5.6	23.1	0.713	0.554	0.910	1.01	0.188	5.47

Architecture Ablation. Table A6 evaluates architectural components and the training strategy. The taxel-topology variant replaces the graph and region-token tactile tokenizer with a flat tactile projection. The local-attention variant removes the temporal locality constraint in visual cross-attention. The remaining variants remove structured attributes, the BPS shape prior, or scheduled teacher forcing. Parallel state generation is evaluated separately in Table A8. Removing taxel topology changes contact F1 from 0.756 to 0.743, pressure vIoU from 0.295 to 0.216, and CoP error from 4.19 to 5.18. This supports the role of the fixed hand-surface layout as a spatial prior for contact and pressure localization. The local visual attention ablation mainly affects contact-frame fingertip accuracy and pressure overlap, suggesting that contact formation and release are best inferred from nearby visual evidence. Removing the structured attributes or BPS prior causes moderate pressure and localization drops, which is consistent with their roles in providing physical and geometric object constraints. Removing teacher forcing produces the largest change in the tactile metrics among the listed variants: contact F1 is 0.713, pressure vIoU is 0.188, and CoP error is 5.47. This pattern is consistent with unreliable pose and contact predictions accumulating through the hierarchy during early optimiza-

Table A7. Weak-context evaluation on UNISON-Sim using the same trained checkpoint without retraining. All rows include RGB. Full context uses per-episode text, attributes, and BPS geometry.

Inference Condition	MPJPE↓	F1↑	vIoU↑
Full context	**18.4**	**0.756**	**0.295**
Text + Attributes	19.3	0.742	0.286
Text + BPS prior	18.8	0.748	0.260
Text (per-episode)	19.7	0.731	0.261
Text (fixed)	20.2	0.720	0.249
Text (empty)	22.1	0.683	0.218

Table A8. Accuracy and efficiency tradeoff on UNISON-Sim. All variants use 30 Euler integration steps. FPS and peak memory are measured per 16-frame clip with batch size 1 on one RTX 5090 GPU. Single-pass denotes parallel state prediction at each flow-time evaluation.

Setting	MPJPE↓	F1↑	vIoU↑	FPS↑	Peak Mem.↓
UNISON	18.4	**0.756**	**0.295**	8.4	11.2 GB
Single-pass generation	19.0	0.737	0.234	18.6	9.1 GB
Flat tactile + Single-pass	19.8	0.715	0.204	**20.1**	**8.7 GB**
Global visual attention	**18.2**	0.748	0.277	5.9	18.4 GB

tion. Scheduled teacher forcing stabilizes the downstream contexts before gradually exposing the model to its own predictions.

Condition Controllability. At inference time, we further evaluate whether UNISON can use non-visual conditions as generation controls. We keep the RGB observation and shape prior fixed, and edit one auxiliary condition at a time, including object mass, sliding friction, and action text. Since the edited attributes define counterfactual interaction settings for the same visual input, this evaluation is qualitative and is used to inspect the controllability of the conditional generator. Figure A8 shows that the generated hand pose remains largely anchored to the input observation across different condition edits. In contrast, the tactile state changes with the edited conditions. Increasing the object mass produces stronger pressure on the main supporting taxels, while lowering the sliding friction changes the pressure distribution toward a different stabilizing pattern. When the action text is changed from "grasp and lift the mug" to "pinch the mug handle before lifting", the generated interaction becomes more localized around the handle-related contact region. These results indicate that UNISON uses physical attributes and action text to modulate the generated tactile state, while preserving the pose geometry supported by the visual input.

Robustness to Weak Object Context. Table A7 evaluates the robustness of UNISON under reduced object context using the same trained checkpoint without retraining. We first remove individual object conditions while retaining the original per-episode text. Without BPS, pressure vIoU decreases from 0.295 to 0.286, while removing attributes reduces it to 0.260. Using only per-episode text with null attributes and BPS yields an MPJPE of 19.7, contact F1 of 0.731, and pressure vIoU of 0.261. We further evaluate settings where informative text is unavailable. The fixed-text setting replaces all episode descriptions with "A hand is grasping an object.", while the empty-text setting passes an empty string through the UMT5 encoder. Both use null attributes and BPS. The fixed-text setting achieves a pressure vIoU of 0.249, which further decreases to 0.218 with empty text. These results show that UNISON remains effective with limited object context, while informative conditions improve generation quality.

Accuracy-Efficiency Tradeoff. The efficiency study evaluates variants that change both computation and accuracy. Table A8 compares the default UNISON model with three alternatives. Single-pass generation removes the staged pose-contact-pressure hierarchy and predicts all state components in parallel at each flow-time or ODE evaluation. All variants use 30 Euler steps. The hierarchical model uses three ordered State DiT passes per step (90 in total), while the parallel variants use one per step (30 in total). Flat tactile + Single-pass further replaces the topology-aware tactile tokens with a flat tactile representation. Global visual attention replaces the local temporal window with attention over all frames. The results show a clear accuracy-efficiency tradeoff. Single-pass generation raises FPS from 8.4 to 18.6 and reduces memory from 11.2 GB to 9.1 GB, while pressure vIoU decreases from 0.295 to 0.234. Combining single-pass generation with flat tactile states gives the highest throughput and lowest memory cost, but further degrades pose accuracy and tactile localization. This confirms that both hierarchical generation and taxel-topology tokenization contribute to the state quality. Global visual attention is slower and more memory intensive than the default model. It also reduces contact and pressure accuracy because the grasp state at each timestep is mainly determined by nearby motion cues, while distant frames can introduce irrelevant visual evidence. The default setting therefore provides the best balance between unified-state accuracy and inference efficiency.

Additional Visualization Results. Figure A9 provides additional qualitative comparisons, while Figure A10 and Figure A11 show representative inference outputs from UNISON. These visualizations provide a broader view of the model's behavior across samples and evaluation settings.

Failure Analysis. Figures A12–A15 summarize four representative failure cases. In Figure A12, the ground truth contains palm contact and pressure, whereas UNISON leaves the corresponding palm taxels inactive. Grasps with palm loading are relatively rare in the training data, and adding such episodes may improve coverage of this pattern. Figure A13 shows an occluded grasp in which the index, middle, and ring fingers are hidden by the object. The ground truth uses a grasp supported by the thumb and index finger, while UNISON generates a different but physically plausible grasp supported by the thumb, index, middle, and ring fingers. This case illustrates the ambiguity of inferring a unique grasp from incomplete visual evidence. In the two episodes in Figure A14, the palm completely occludes the object, leading the generated contact and pressure patterns to differ substantially from the ground truth. Figure A15 evaluates a failed grasp. After the object is dropped, the generated thumb contact persists briefly before disappearing. Because all 45,124 retained UNISON-Sim episodes depict successful grasps, release dynamics after grasp failure fall outside the training distribution. Adding failed episodes with explicit contact release transitions may improve the timing of contact termination.

H. Limitations and Future Work

UNISON still relies primarily on simulation for complete supervision of pose, contact, and pressure. Real-world adaptation updates only the visual encoder while keeping the state generator frozen, and the real validation set lacks hand pose ground truth. Consequently, contact and pressure can be evaluated against direct measurements, whereas real-world pose quality is assessed only through agreement with a WiLoR pseudo-reference. The generator also inherits the coverage bias of UNISON-Sim, whose 45,124 retained episodes are all successful grasps. Rare palm loading patterns and failure dynamics involving object slippage, object drops, and delayed contact release are therefore less well represented. Future work will develop a minimally obtrusive acquisition system that synchronizes hand pose with contact and pressure measurements. Thin-film sensors will be placed only on selected palm and finger regions to limit changes to natural hand appearance and motion. We plan to use this system to collect a large real-world dataset whose visual observations remain close to bare-hand interactions. The dataset will include successful grasps as well as underrepresented events such as palm loading, object slippage, object drops, and contact release. Such data will support supervised training and direct evaluation of the unified state generator in real scenes.

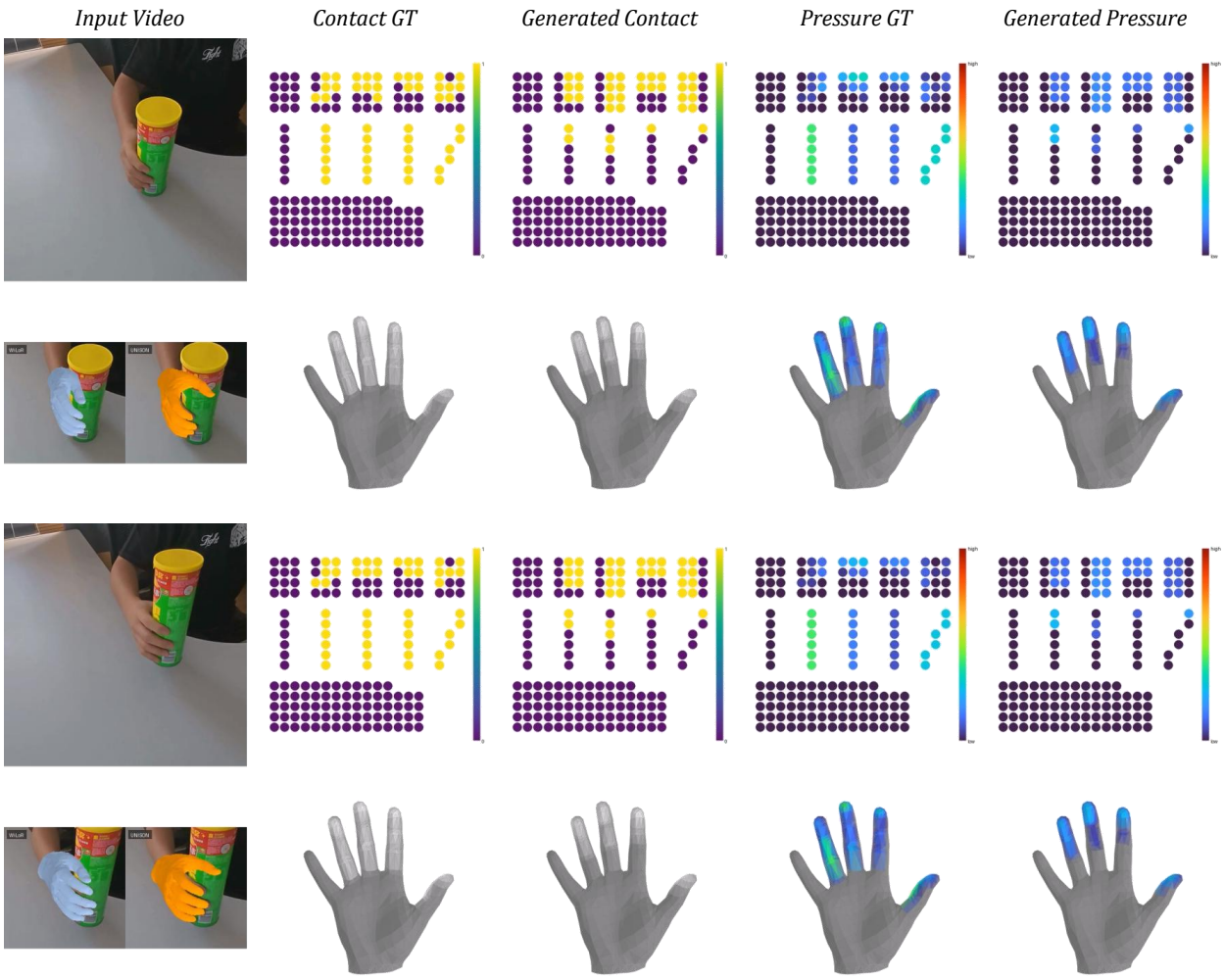


Figure A5. Additional bare-hand results on real-world videos (1/3). Representative frames are shown with the input RGB observation and a MANO hand pose comparison, where the WiLoR reconstruction is displayed on the left and the retargeted UNISON result on the right. The remaining columns show ground-truth and generated contact and pressure on the taxel topology and hand surface. Ground-truth contact and pressure are obtained from synchronized tactile measurements during matched gloved-hand grasps.

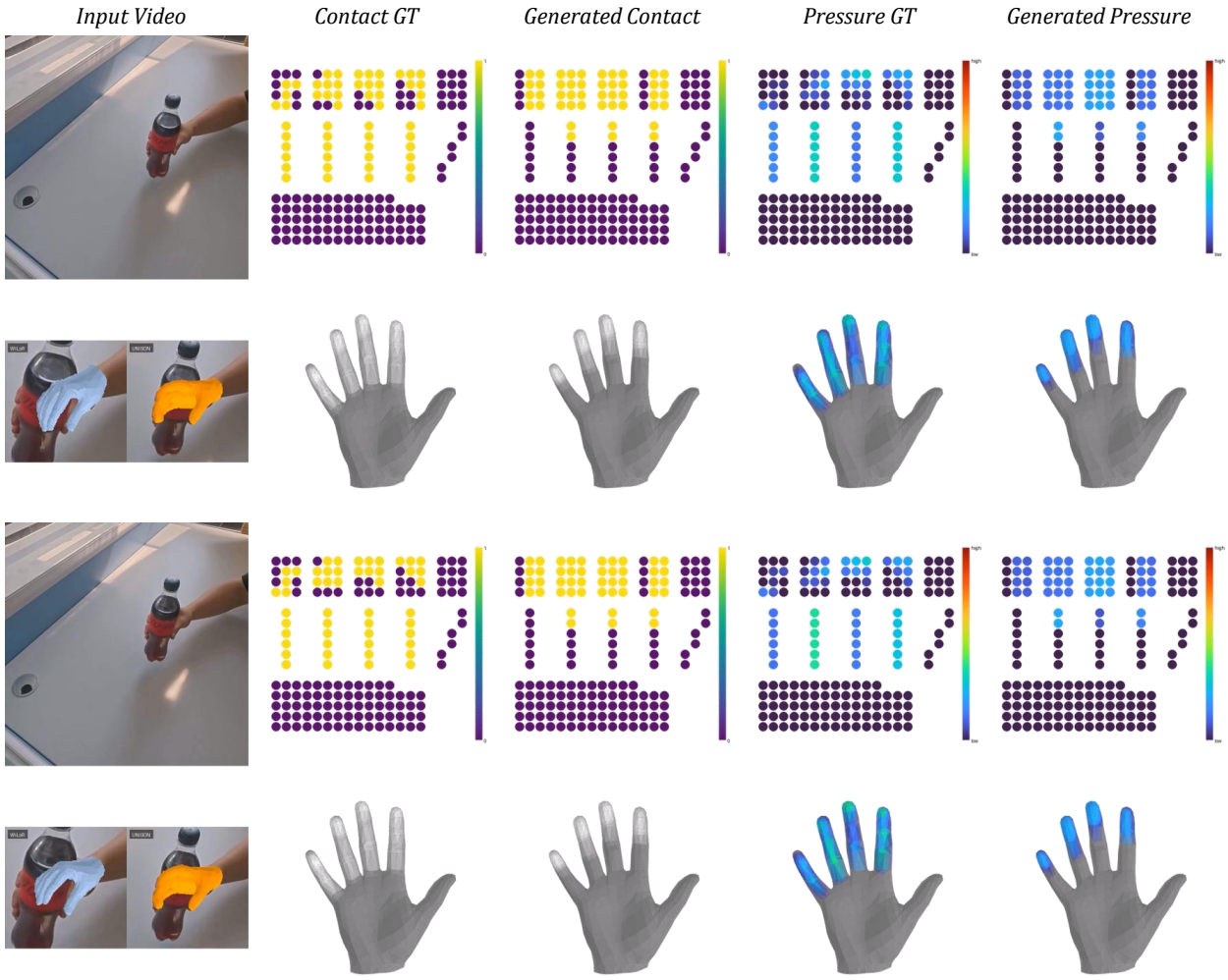


Figure A6. Additional bare-hand results on real-world videos (2/3). Representative frames are shown with the input RGB observation and a MANO hand pose comparison, where the WiLoR reconstruction is displayed on the left and the retargeted UNISON result on the right. The remaining columns show ground-truth and generated contact and pressure on the taxel topology and hand surface. Ground-truth contact and pressure are obtained from synchronized tactile measurements during matched gloved-hand grasps.

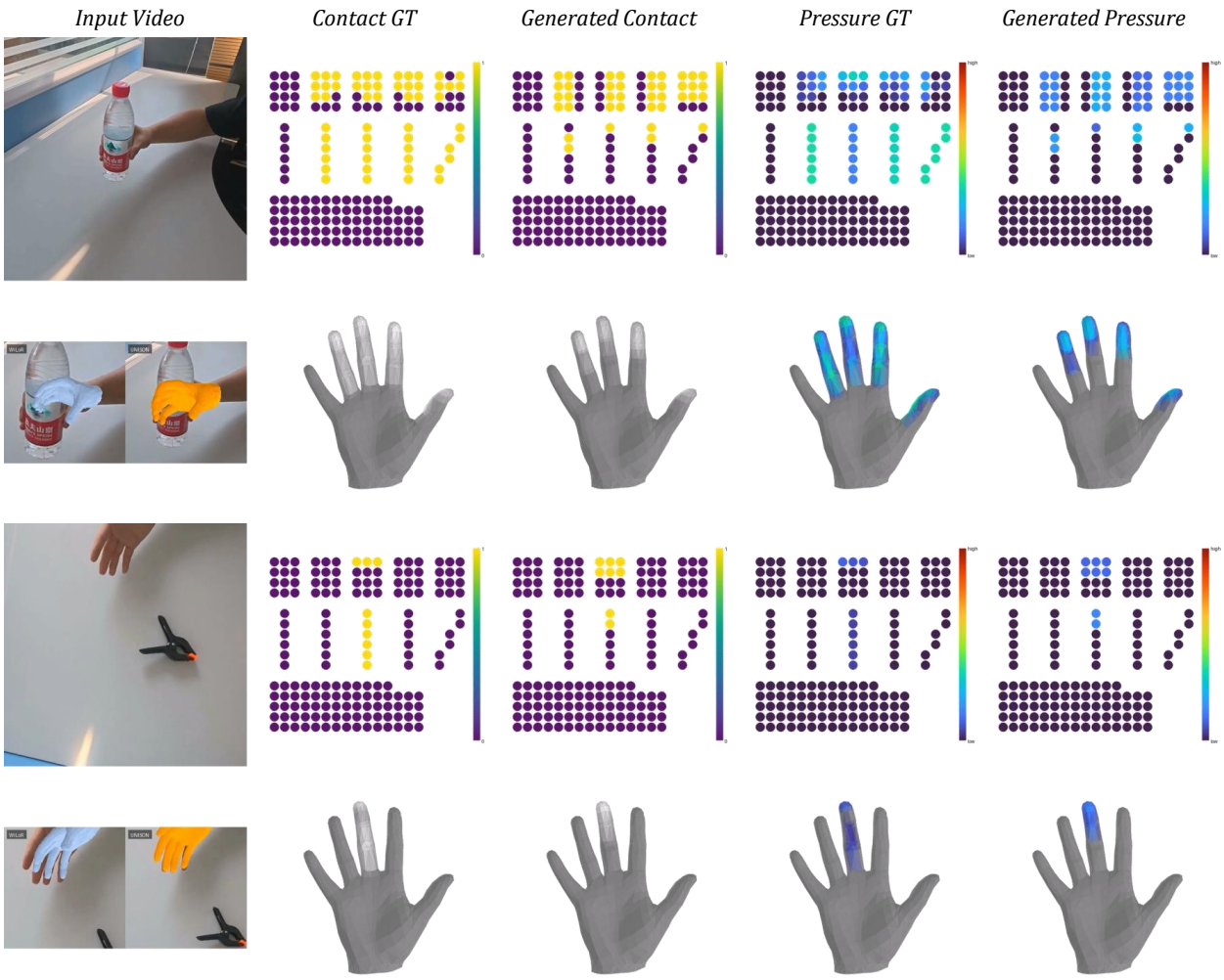


Figure A7. Additional bare-hand results on real-world videos (3/3). Representative frames are shown with the input RGB observation and a MANO hand pose comparison, where the WiLoR reconstruction is displayed on the left and the retargeted UNISON result on the right. The remaining columns show ground-truth and generated contact and pressure on the taxel topology and hand surface. Ground-truth contact and pressure are obtained from synchronized tactile measurements during matched gloved-hand grasps.

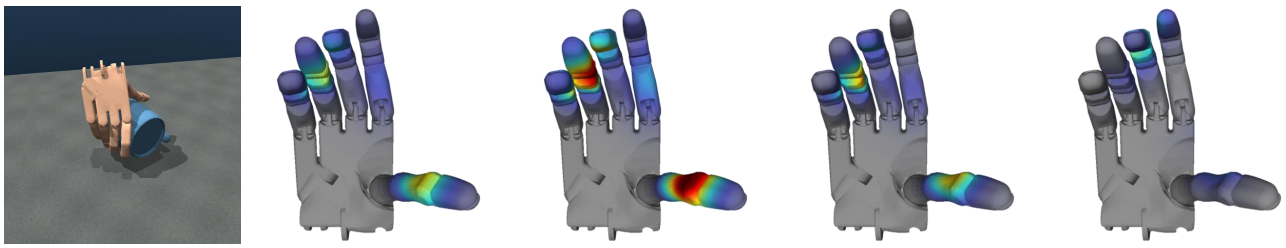


Figure A8. Condition-controlled generation under a fixed RGB observation. The leftmost image shows the input frame, and the four columns on the right show generated interaction states under different conditions. Column (a) is the baseline condition. Columns (b), (c), and (d) apply one counterfactual edit at a time, including heavier object mass, lower sliding friction, and a different action text. Red text marks the edited condition relative to the baseline. The pose geometry remains visually consistent across edits, while the generated tactile state, especially the pressure distribution and magnitude, changes according to the edited condition.

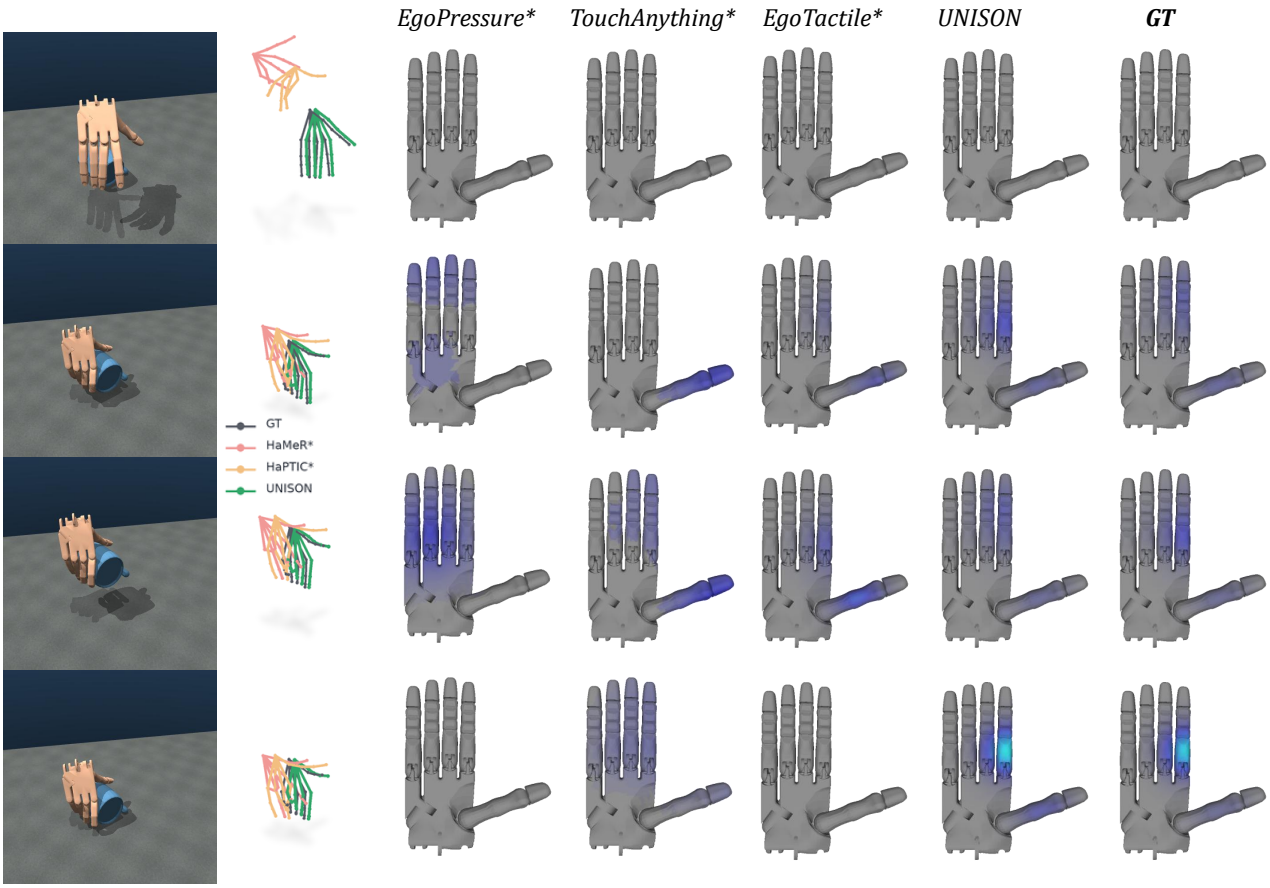


Figure A9. Additional qualitative comparison results.

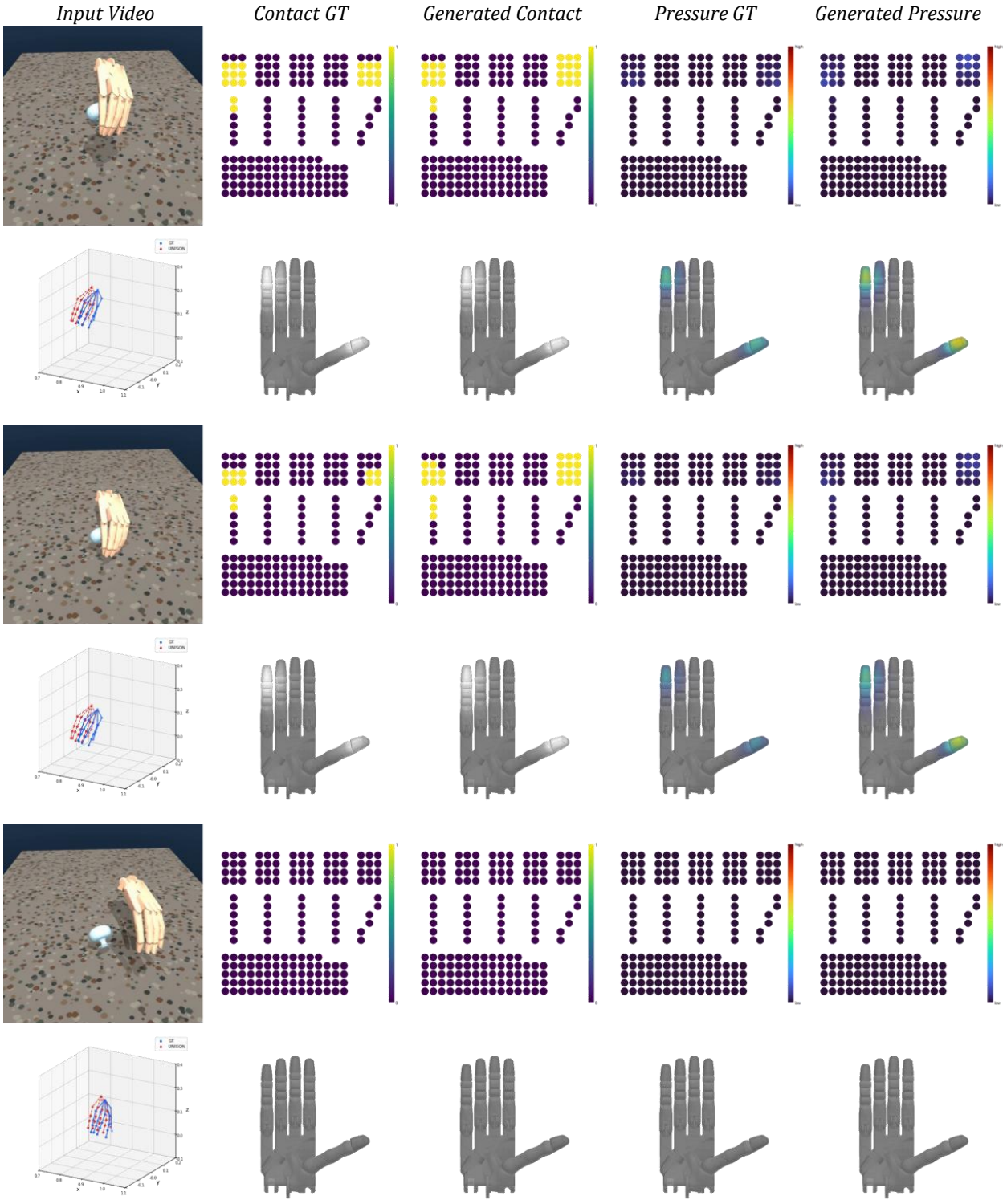


Figure A10. UNISON inference results on representative input videos (1/2). From left to right, each example shows the input video frame, the ground-truth and UNISON-generated contact maps, and the ground-truth and UNISON-generated pressure maps. UNISON accurately recovers the spatial support of hand-object contact and captures the corresponding pressure distribution over the hand surface.

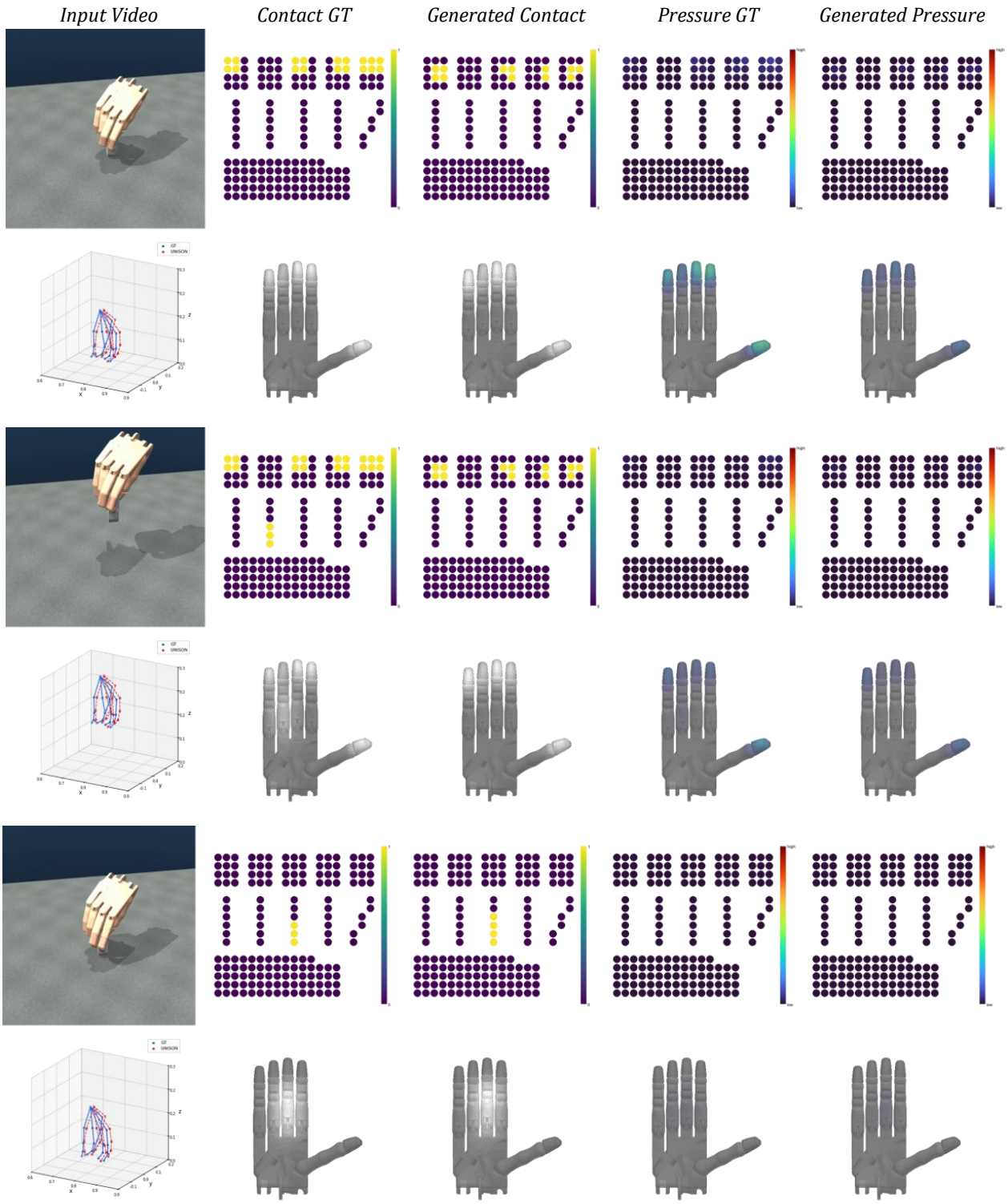


Figure A11. UNISON inference results on representative input videos (2/2). From left to right, each example shows the input video frame, the ground-truth and UNISON-generated contact maps, and the ground-truth and UNISON-generated pressure maps. UNISON accurately recovers the spatial support of hand-object contact and captures the corresponding pressure distribution over the hand surface.

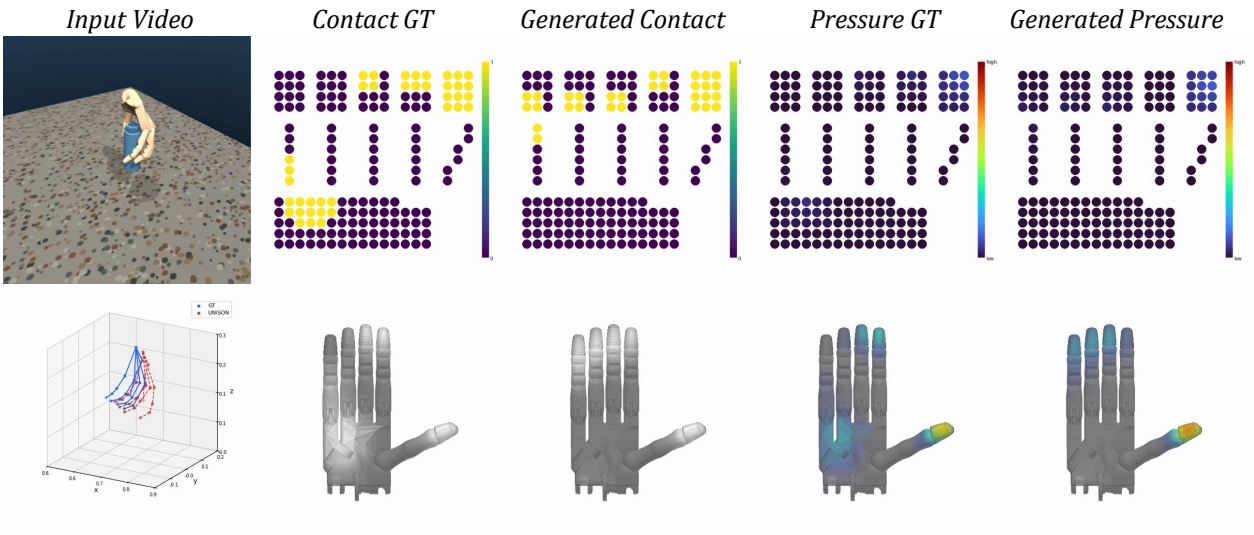


Figure A12. Representative failure cases (1/4). Missed palm loading caused by limited training coverage. The ground truth contains active contact and pressure taxels on the palm, while the UNISON outputs omit the corresponding palm response. This case represents grasp configurations in which the palm directly supports the object, a pattern that occurs relatively infrequently in the training data.

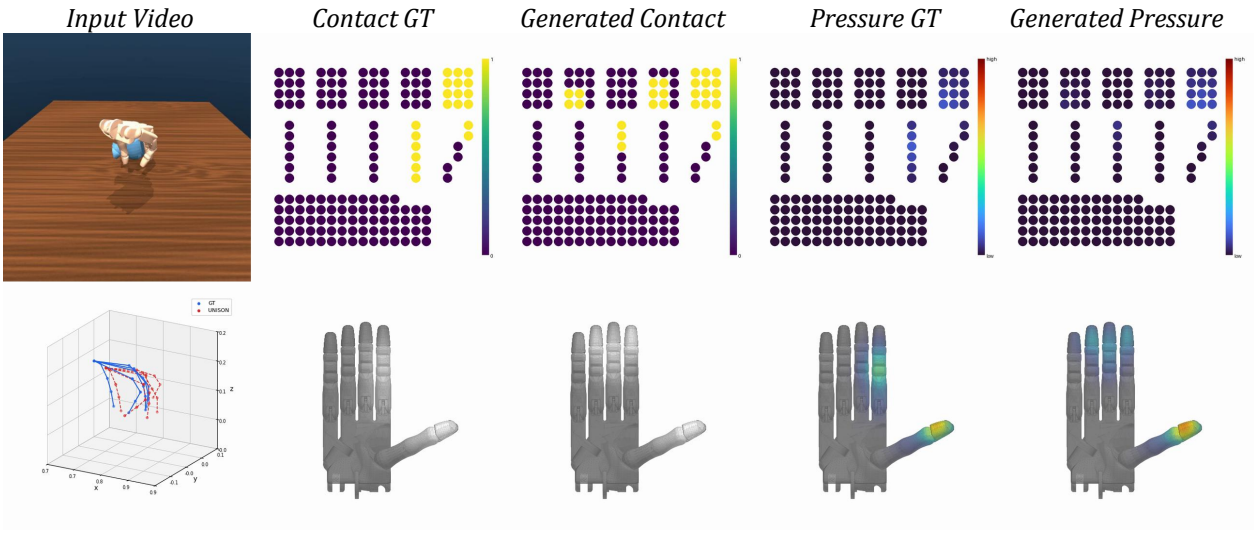


Figure A13. Representative failure cases (2/4). Alternative grasp generation under severe finger occlusion. The object occludes the index, middle, and ring fingers. The ground truth uses a grasp supported by the thumb and index finger, whereas UNISON generates a plausible alternative supported by the thumb, index, middle, and ring fingers. The visibly unloaded little finger remains inactive.

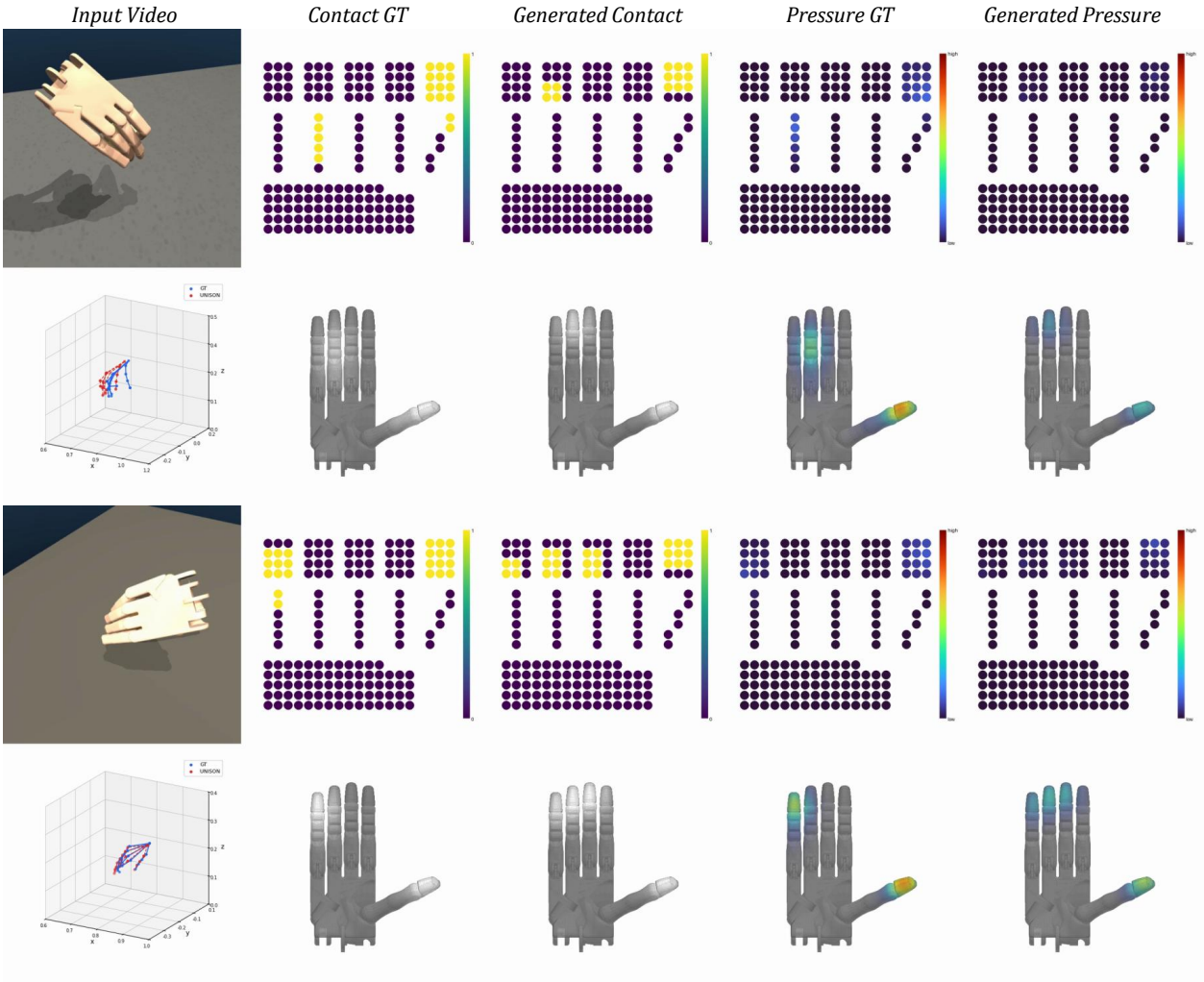


Figure A14. Representative failure cases (3/4). Tactile state ambiguity under complete object occlusion. In both grasping episodes, the palm fully occludes the object in the input view. The absence of visible geometry between the hand and object leads the contact and pressure maps generated by UNISON to differ substantially from the ground truth.

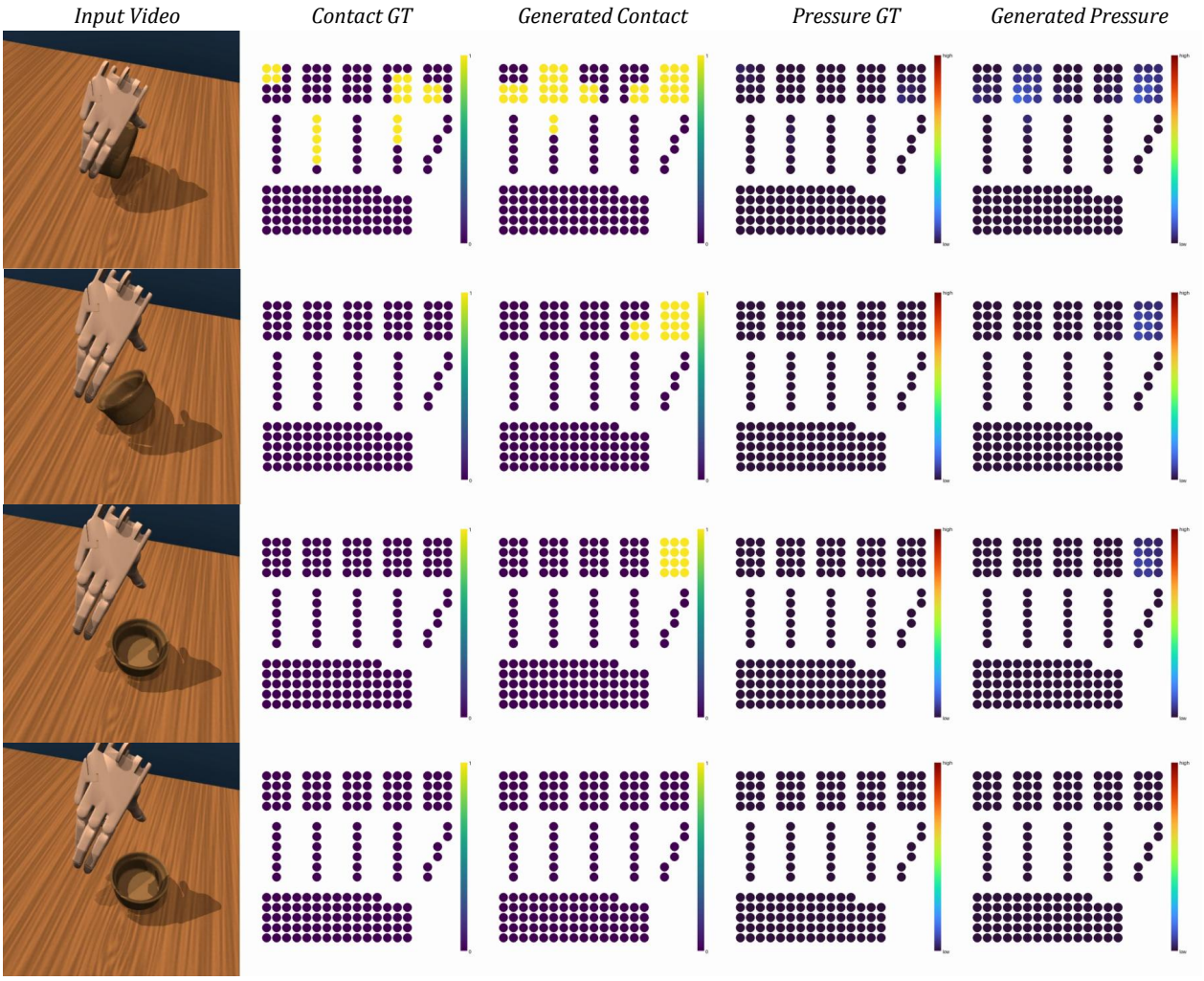


Figure A15. Representative failure cases (4/4). Delayed contact release after a failed grasp. The object drops from the hand, but the generated thumb contact remains active for several subsequent frames before disappearing. Because failed interactions are absent from the training distribution, this temporal lag reveals limited modeling of contact termination after grasp failure.